PRACTICAL 1
Installing RHEL and Studying the Linux Graphical Environment

Aim

To install Red Hat Enterprise Linux in VirtualBox/VMware and study the graphical environment, login manager, desktop environment, directory structure and file systems.

Requirements

- VirtualBox or VMware
- RHEL ISO
- 4 GB RAM or more
- 30 GB virtual disk

A. Creating the Virtual Machine

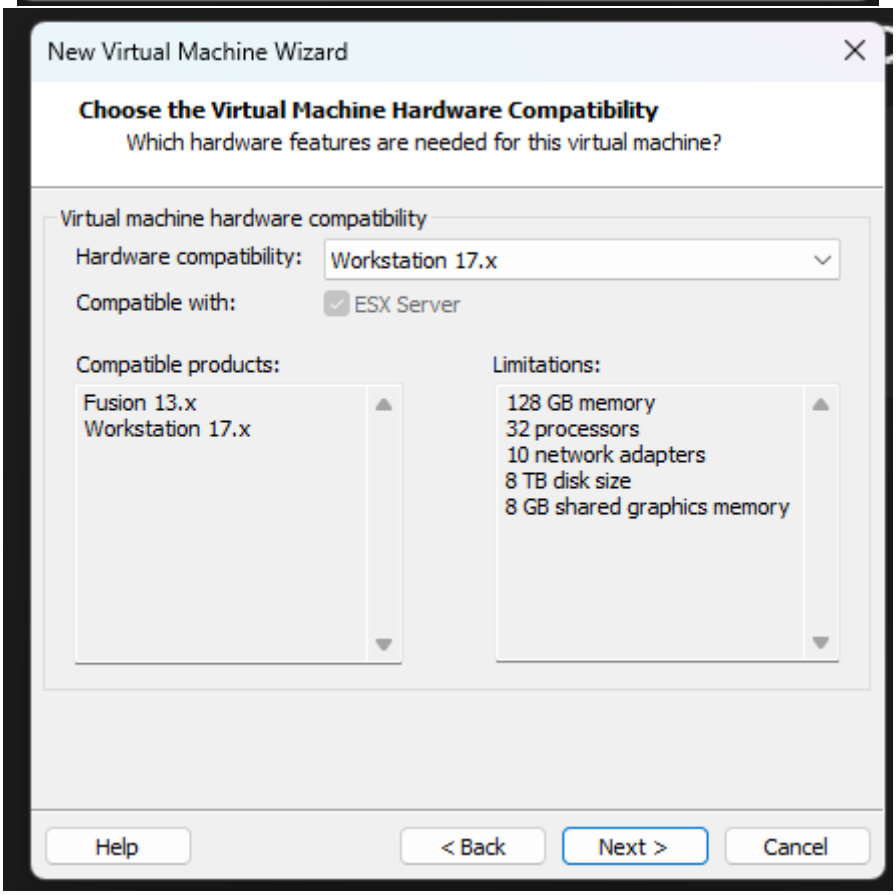
Procedure

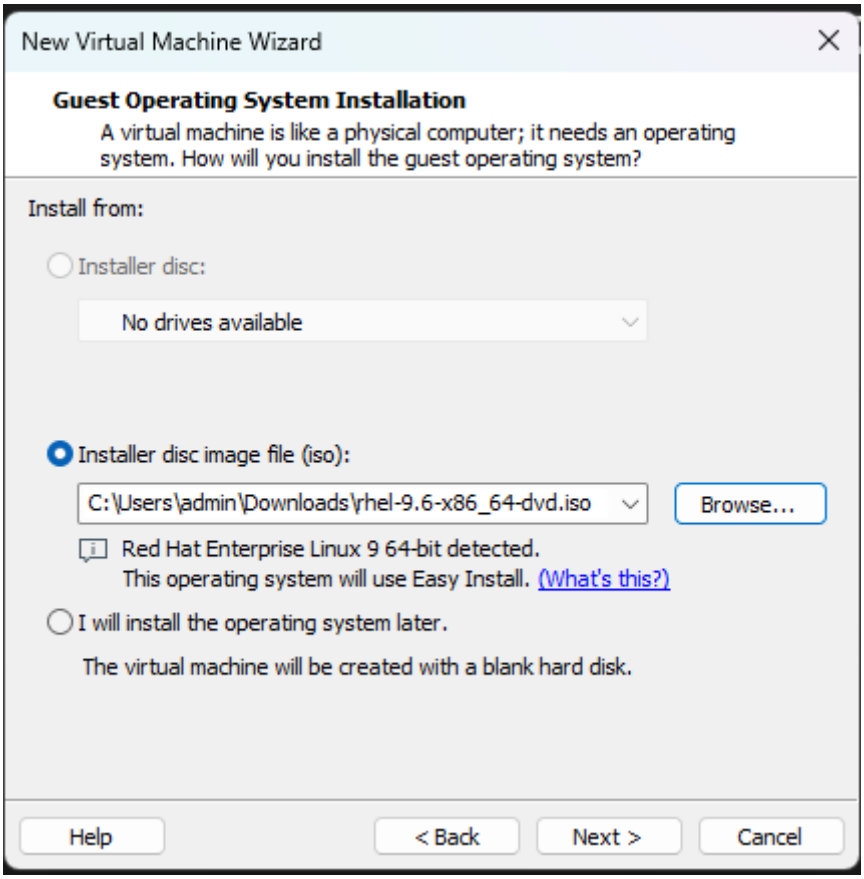
1. Open VirtualBox/VMware.
2. Create a new virtual machine.
3. Select Linux / Red Hat Enterprise Linux 64-bit.
4. Allocate RAM.
5. Create a virtual hard disk.
6. Attach the RHEL ISO.
7. Start the VM.
8. Select **Install Red Hat Enterprise Linux**.
9. Select language and keyboard.
10. Configure:

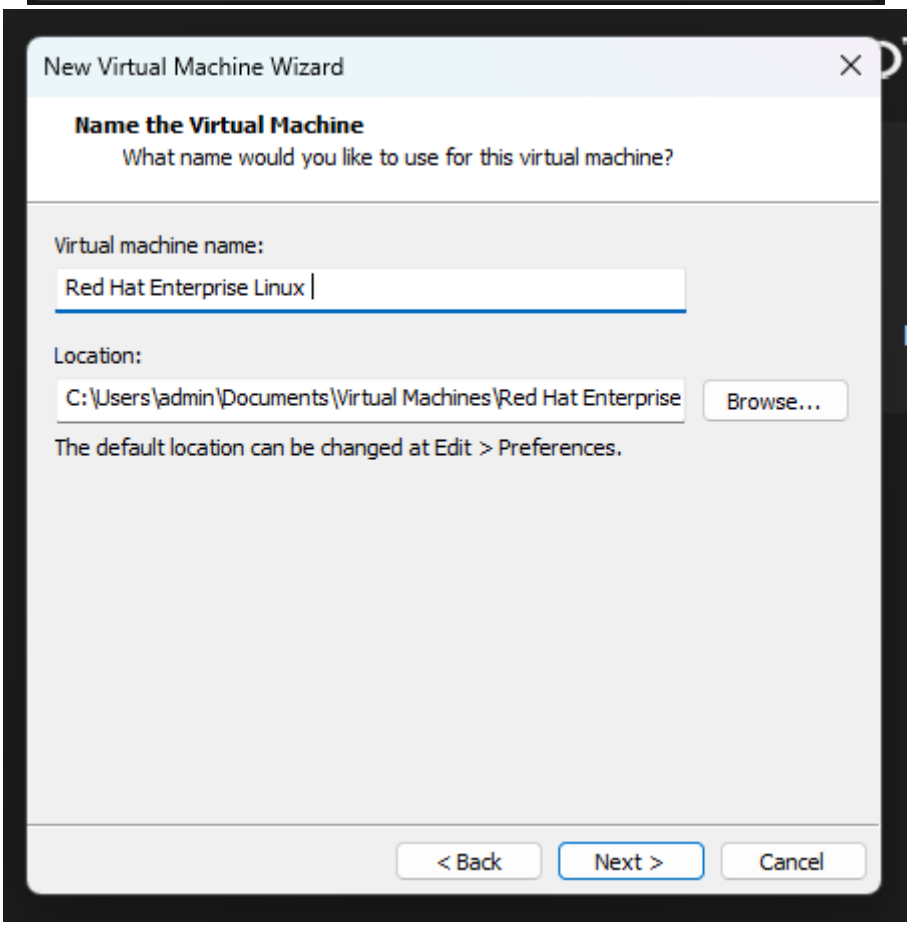
- Installation destination
- Network
- Time zone
- User account
- Root password

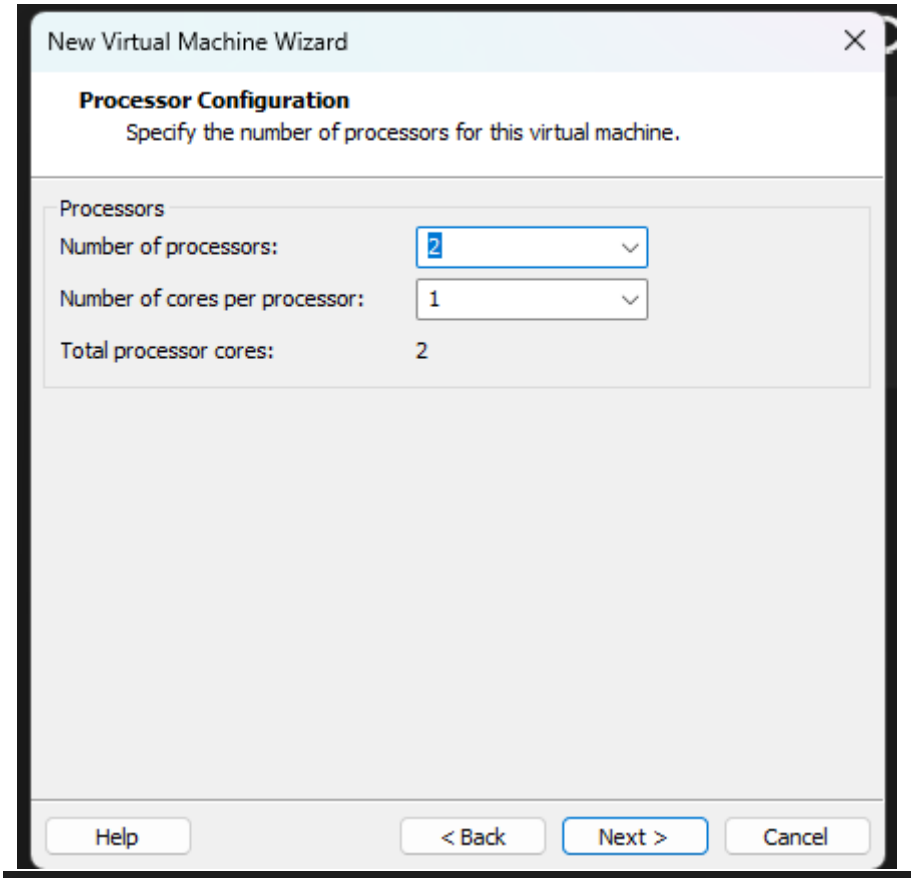
11. Begin installation.
12. Reboot after installation.
13. Log in.

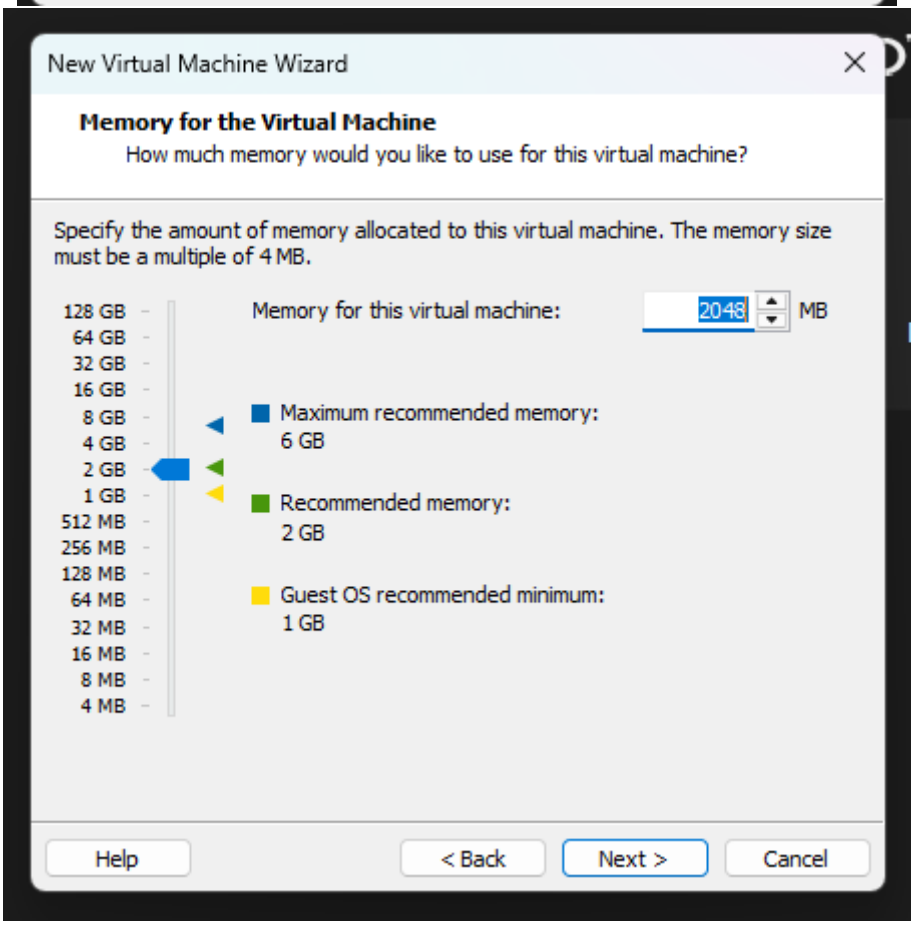

New Virtual Machine Wizard
vmware
WORKSTATION
PRO™
17
Welcome to the New Virtual Machine Wizard
What type of configuration do you want?
Typical (recommended)
Create a Workstation 17.x virtual machine in a few easy steps.
Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.
Help
< Back
Next >
Cancel

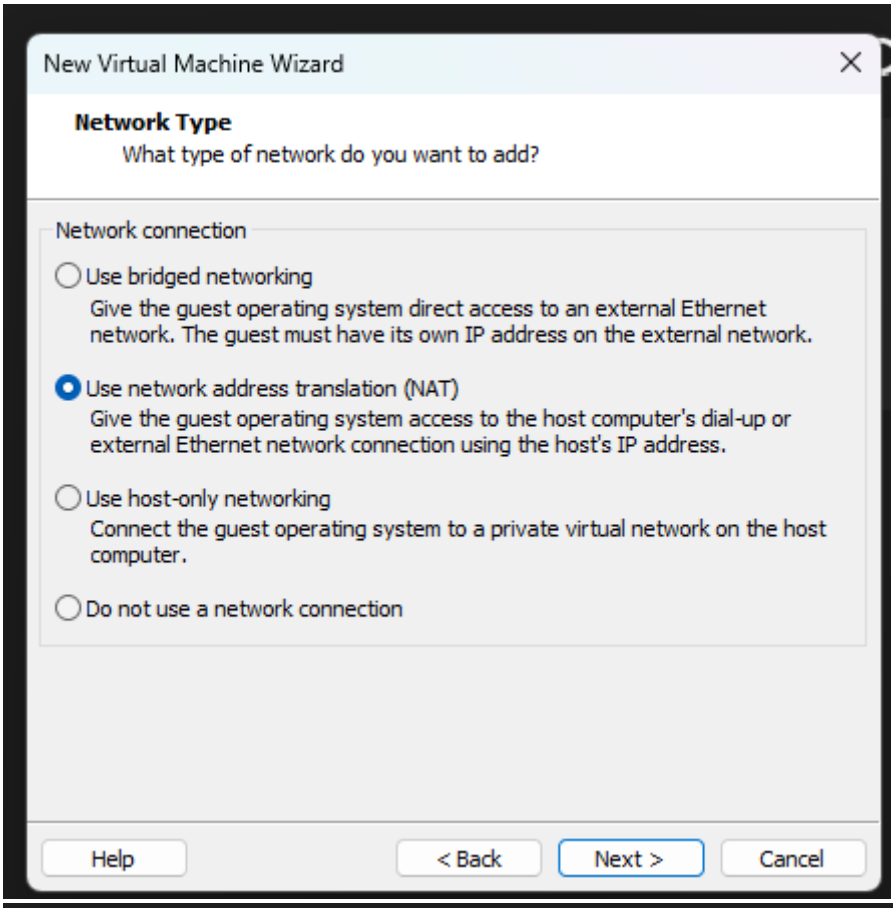

New Virtual Machine Wizard
Choose the Virtual Machine Hardware Compatibility
Which hardware features are needed for this virtual machine?
Virtual machine hardware compatibility
Hardware compatibility:
Workstation 17.x
Compatible with:
ESX Server
Compatible products:
Fusion 13.x
Workstation 17.x
Limitations:
128 GB memory
32 processors
10 network adapters
8 TB disk size
8 GB shared graphics memory
Help
< Back
Next >
Cancel

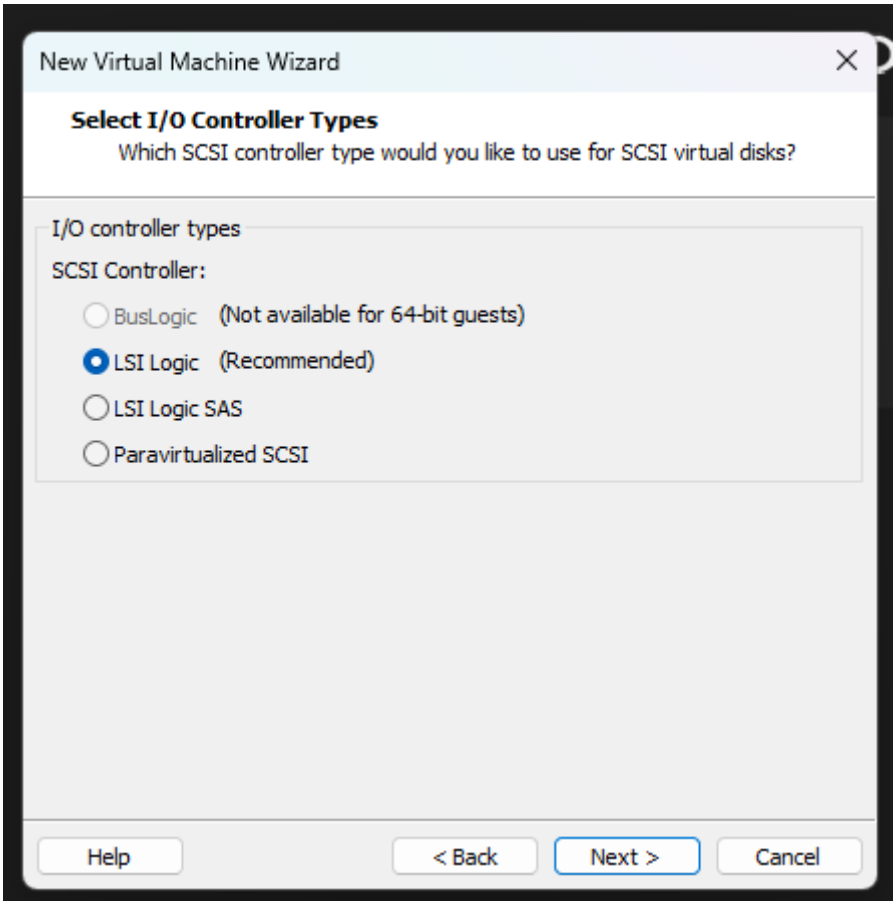

New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
C:\Users\admin\Downloads\rhel-9.6-x86_64-dvd.iso
Browse...
Red Hat Enterprise Linux 9 64-bit detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel

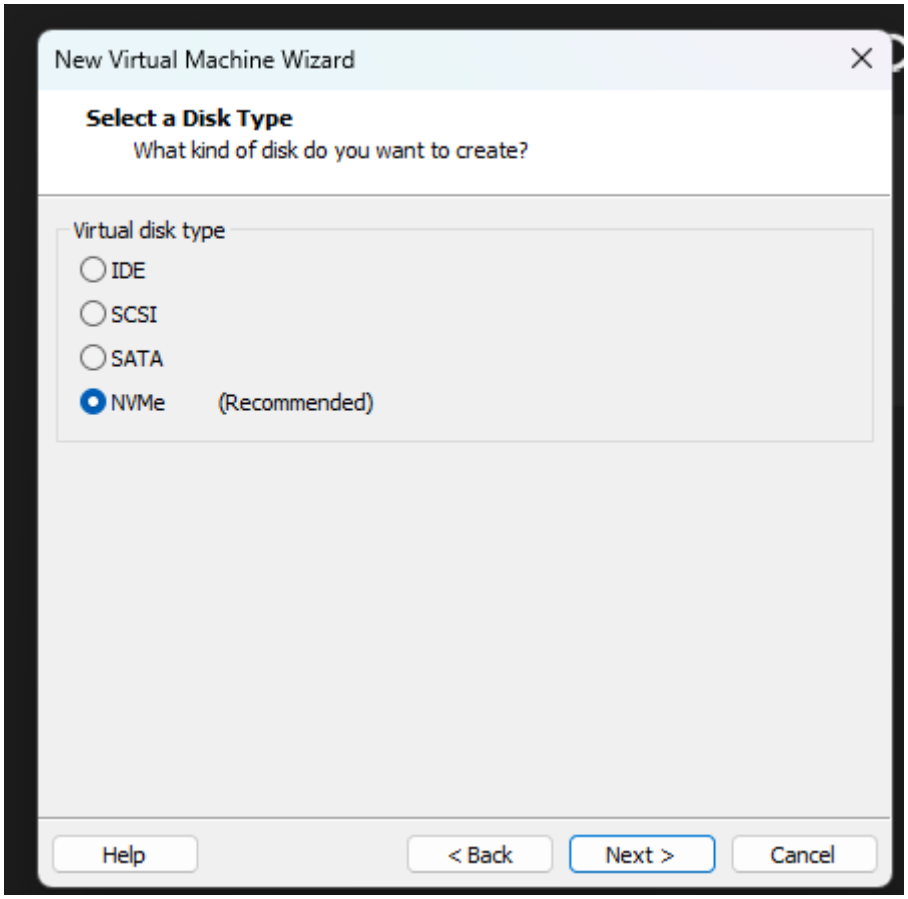

New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Red Hat Enterprise Linux
Location:
C:\Users\admin\Documents\Virtual Machines\Red Hat Enterprise
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel

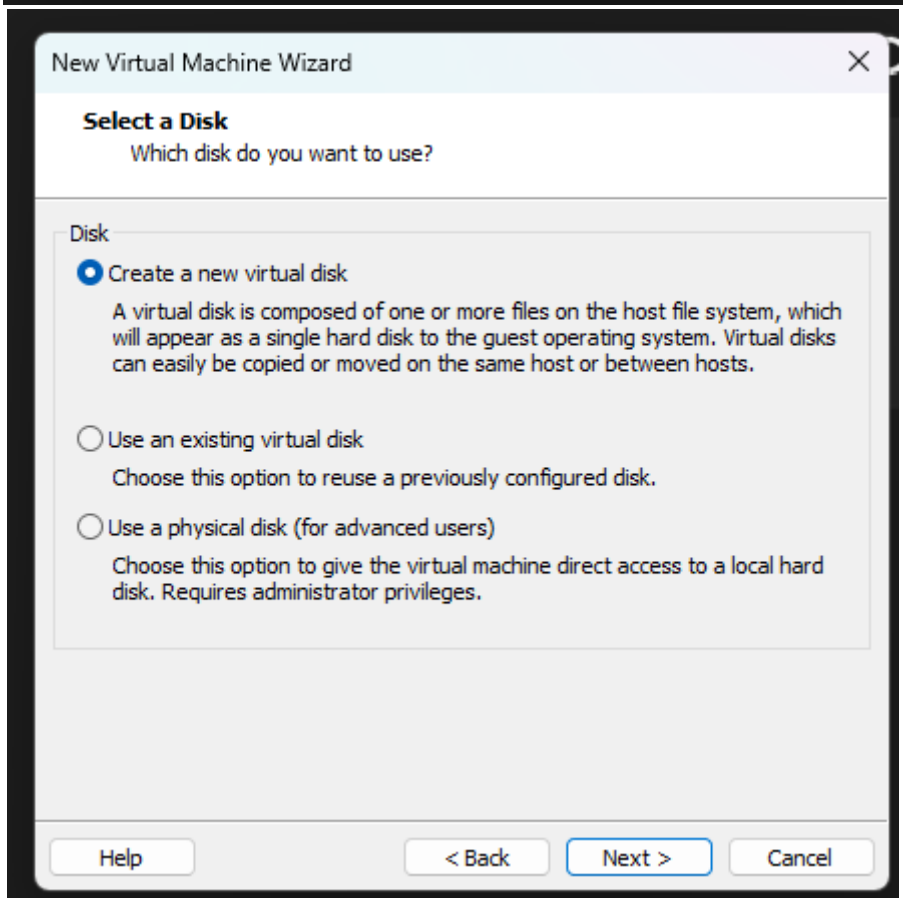

New Virtual Machine Wizard
Processor Configuration
Specify the number of processors for this virtual machine.
Processors
Number of processors:
2
Number of cores per processor:
1
Total processor cores:
2
Help
< Back
Next >
Cancel

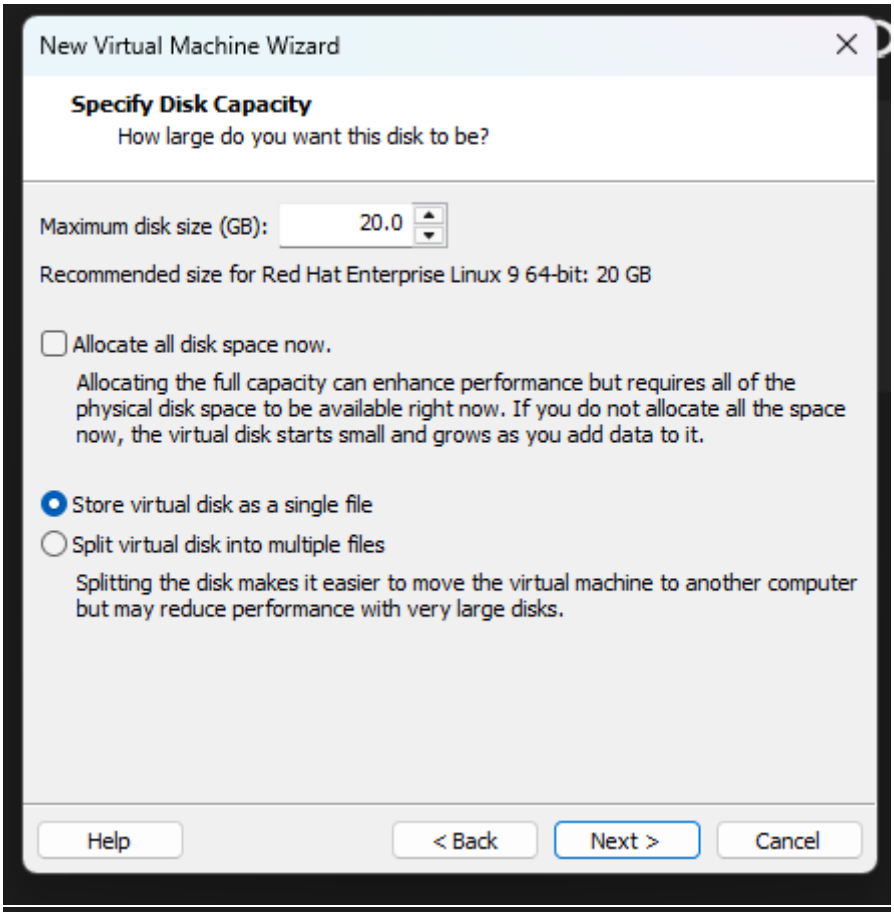

New Virtual Machine Wizard
Memory for the Virtual Machine
How much memory would you like to use for this virtual machine?
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Memory for this virtual machine:
2048
MB
Maximum recommended memory:
6 GB
Recommended memory:
2 GB
Guest OS recommended minimum:
1 GB
Help
< Back
Next >
Cancel

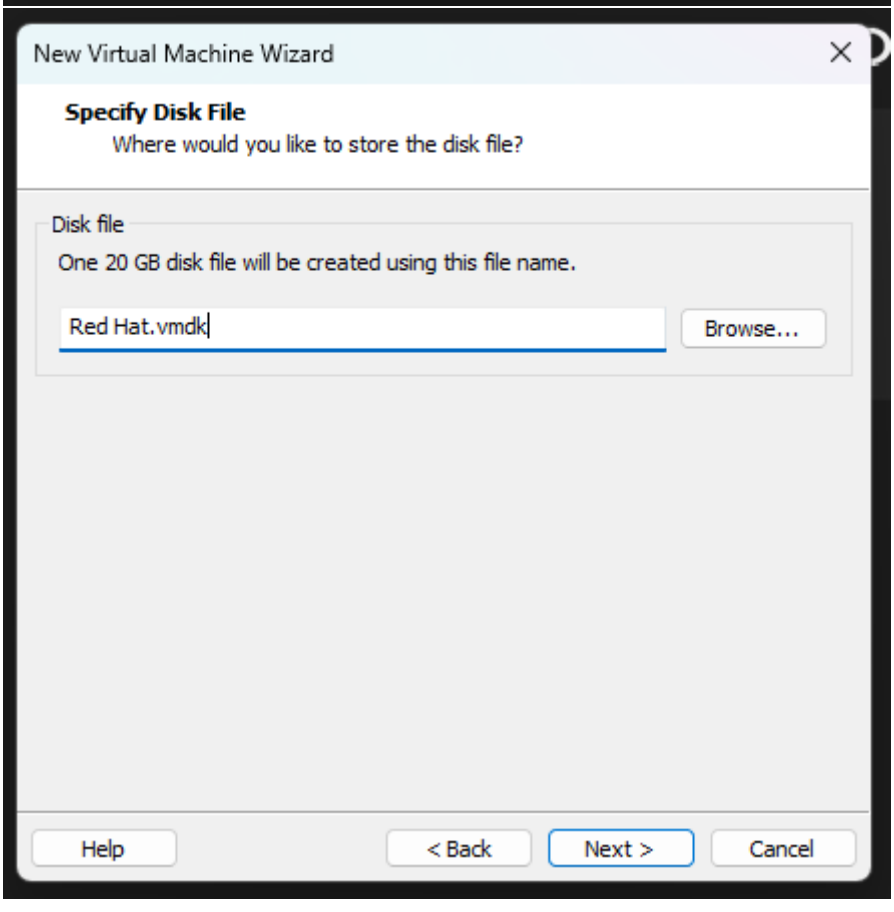

New Virtual Machine Wizard
Network Type
What type of network do you want to add?
Network connection
Use bridged networking
Give the guest operating system direct access to an external Ethernet network. The guest must have its own IP address on the external network.
Use network address translation (NAT)
Give the guest operating system access to the host computer's dial-up or external Ethernet network connection using the host's IP address.
Use host-only networking
Connect the guest operating system to a private virtual network on the host computer.
Do not use a network connection
Help
< Back
Next >
Cancel

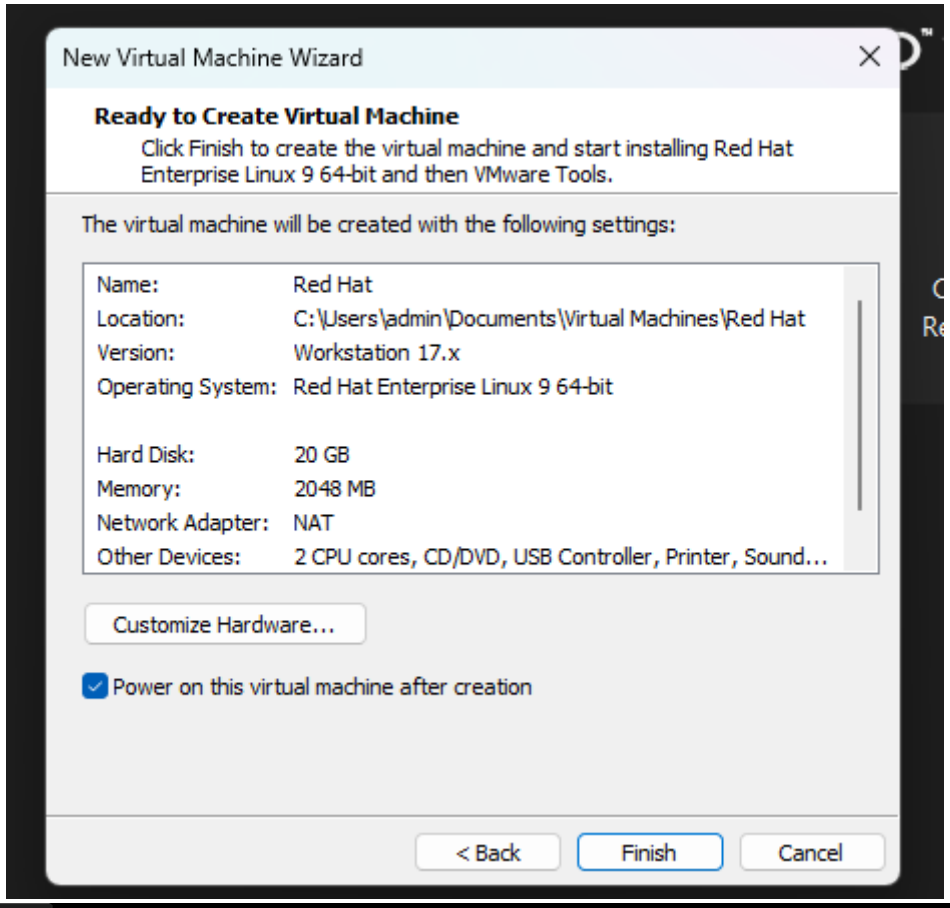

New Virtual Machine Wizard
Select I/O Controller Types
Which SCSI controller type would you like to use for SCSI virtual disks?
I/O controller types
SCSI Controller:
BusLogic (Not available for 64-bit guests)
LSI Logic (Recommended)
LSI Logic SAS
Paravirtualized SCSI
Help
< Back
Next >
Cancel

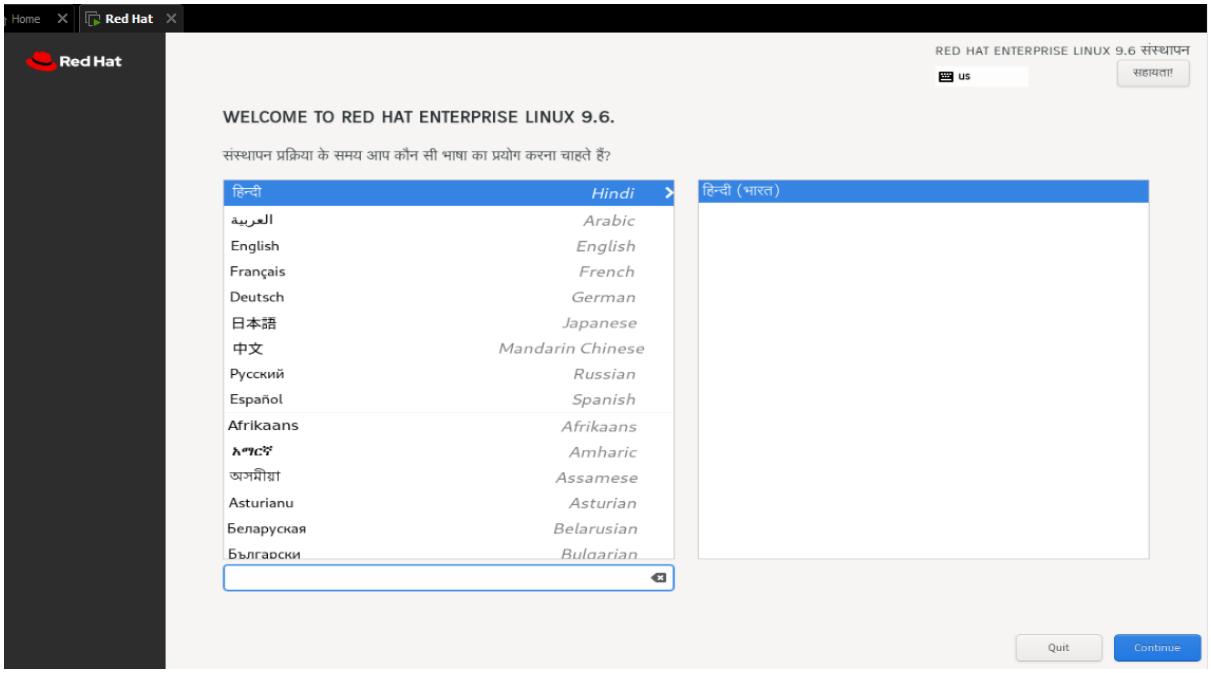

New Virtual Machine Wizard
Select a Disk Type
What kind of disk do you want to create?
Virtual disk type
IDE
SCSI
SATA
NVMe (Recommended)
Help
< Back
Next >
Cancel

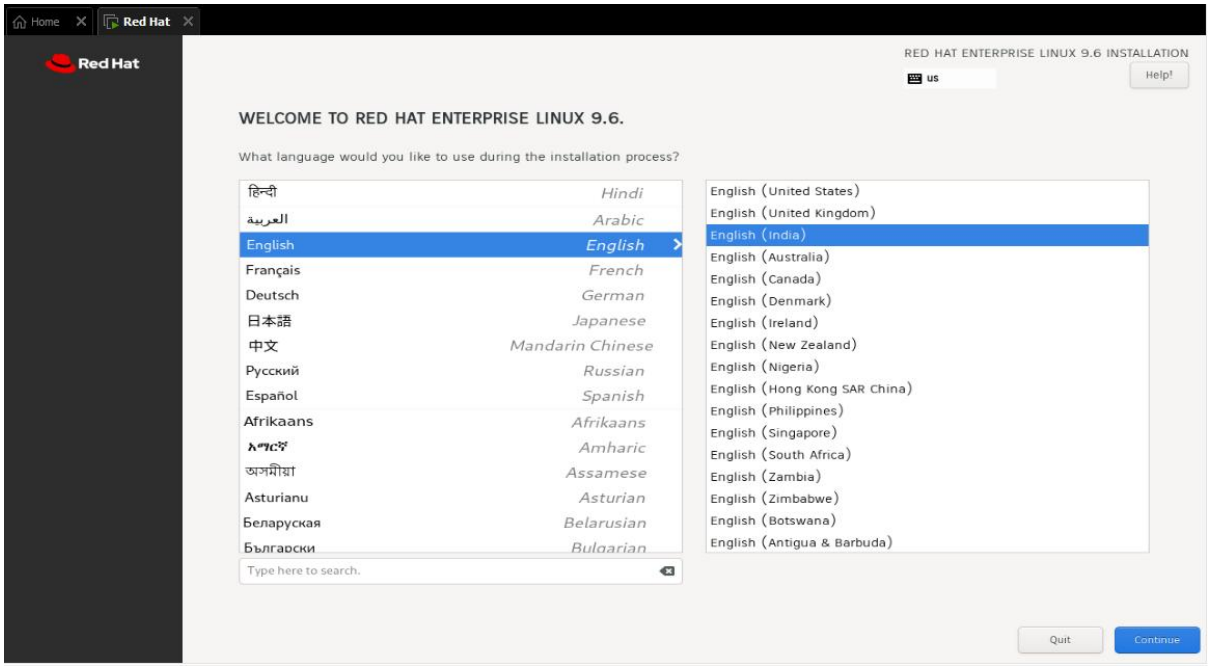

New Virtual Machine Wizard
Select a Disk
Which disk do you want to use?
Disk
Create a new virtual disk
A virtual disk is composed of one or more files on the host file system, which will appear as a single hard disk to the guest operating system. Virtual disks can easily be copied or moved on the same host or between hosts.
Use an existing virtual disk
Choose this option to reuse a previously configured disk.
Use a physical disk (for advanced users)
Choose this option to give the virtual machine direct access to a local hard disk. Requires administrator privileges.
Help
< Back
Next >
Cancel

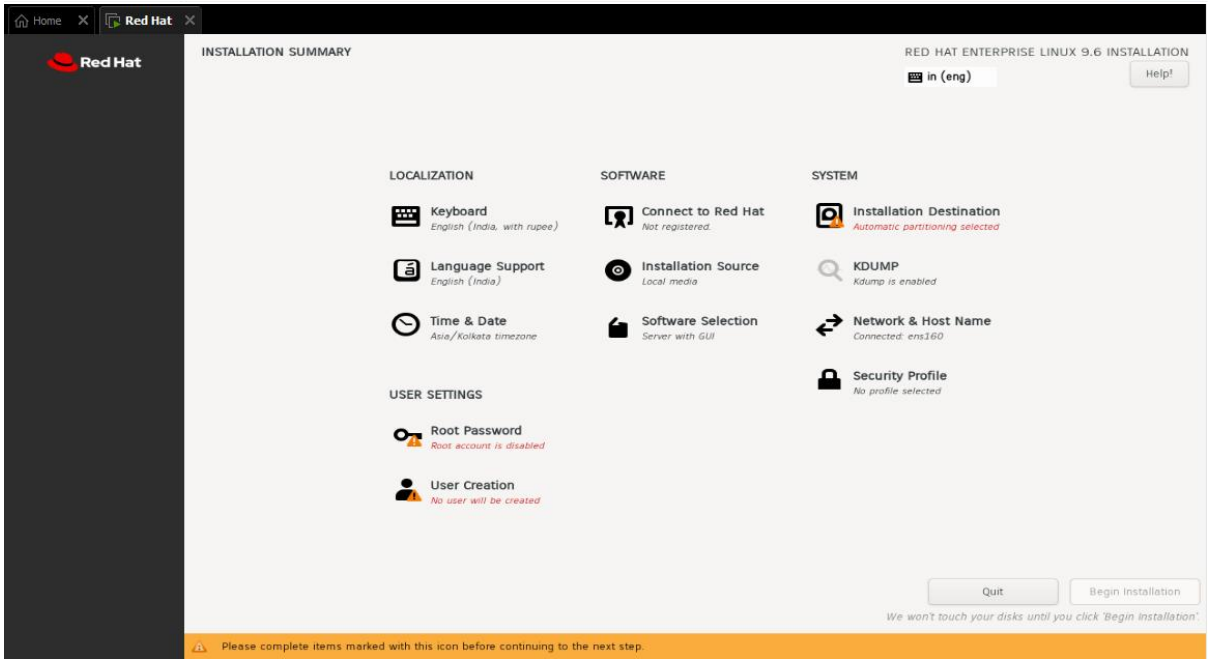

New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
Maximum disk size (GB):
20.0
Recommended size for Red Hat Enterprise Linux 9 64-bit: 20 GB
Allocate all disk space now.
Allocating the full capacity can enhance performance but requires all of the physical disk space to be available right now. If you do not allocate all the space now, the virtual disk starts small and grows as you add data to it.
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel

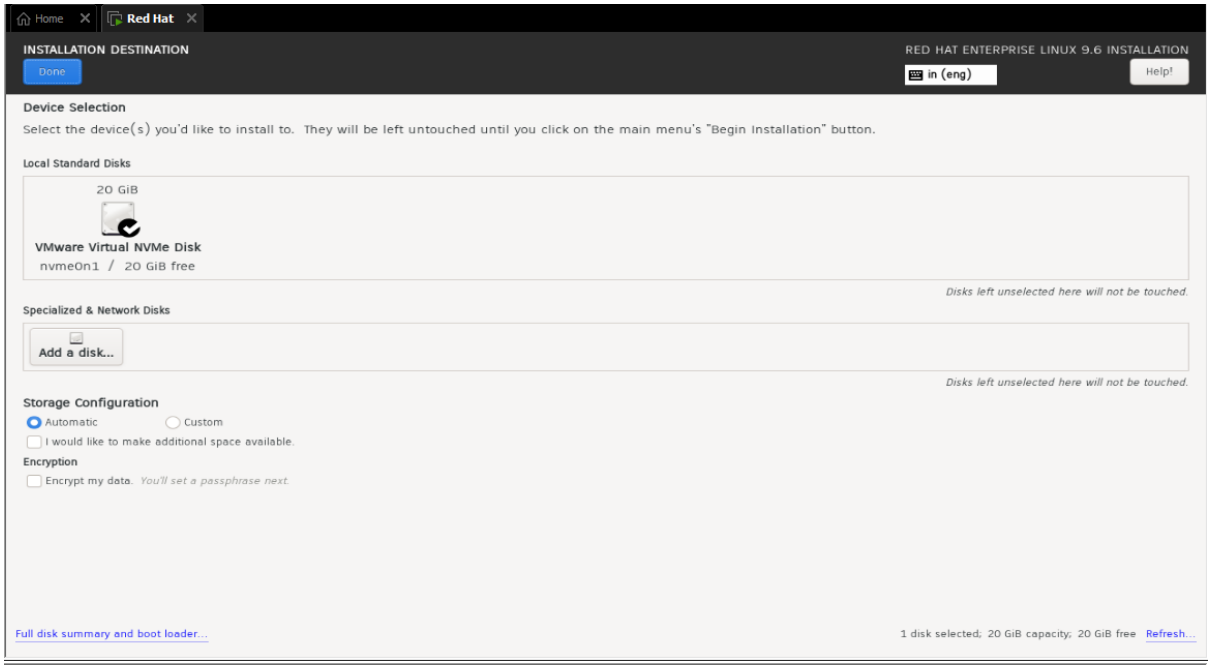

New Virtual Machine Wizard
Specify Disk File
Where would you like to store the disk file?
Disk file
One 20 GB disk file will be created using this file name.
Red Hat.vmdk
Browse...
Help
< Back
Next >
Cancel

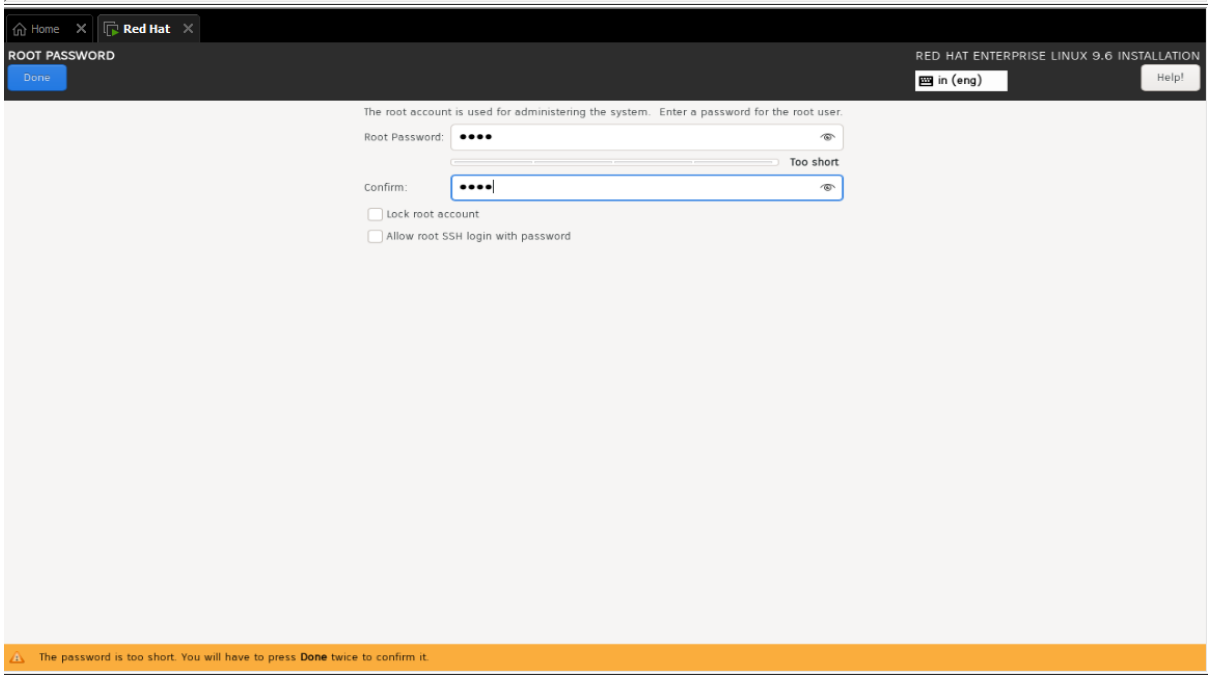

New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Red Hat Enterprise Linux 9 64-bit and then VMware Tools.
The virtual machine will be created with the following settings:
Name: Red Hat
Location: C:\Users\admin\Documents\Virtual Machines\Red Hat
Version: Workstation 17.x
Operating System: Red Hat Enterprise Linux 9 64-bit
Hard Disk: 20 GB
Memory: 2048 MB
Network Adapter: NAT
Other Devices: 2 CPU cores, CD/DVD, USB Controller, Printer, Sound...
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel

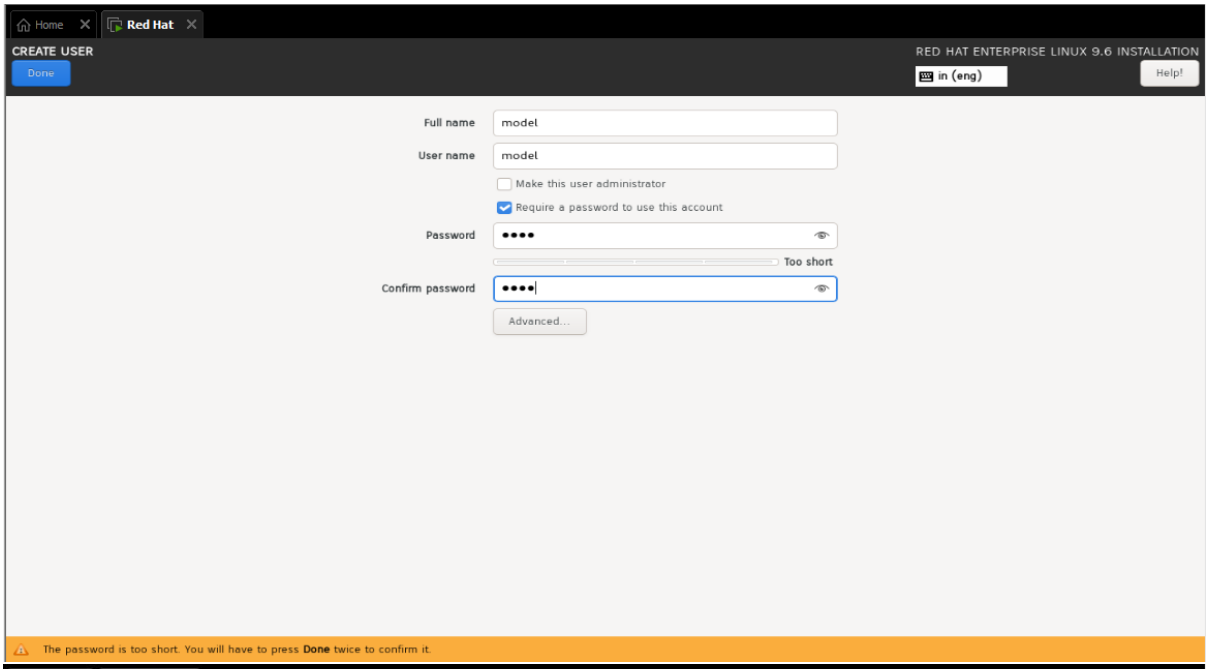

Home
Red Hat
Red Hat
RED HAT ENTERPRISE LINUX 9.6 संस्थापन
us
सहायता!
WELCOME TO RED HAT ENTERPRISE LINUX 9.6.
संस्थापन प्रक्रिया के समय आप कौन सी भाषा का प्रयोग करना चाहते हैं?
हिन्दी Hindi
العربية Arabic
English English
Français French
Deutsch German
日本語 Japanese
中文 Mandarin Chinese
Русский Russian
Español Spanish
Afrikaans Afrikaans
አማርኛ Amharic
অসমীয়া Assamese
Asturianu Asturian
Беларуская Belarusian
Български Bulgarian
हिन्दी (भारत)
Quit
Continue

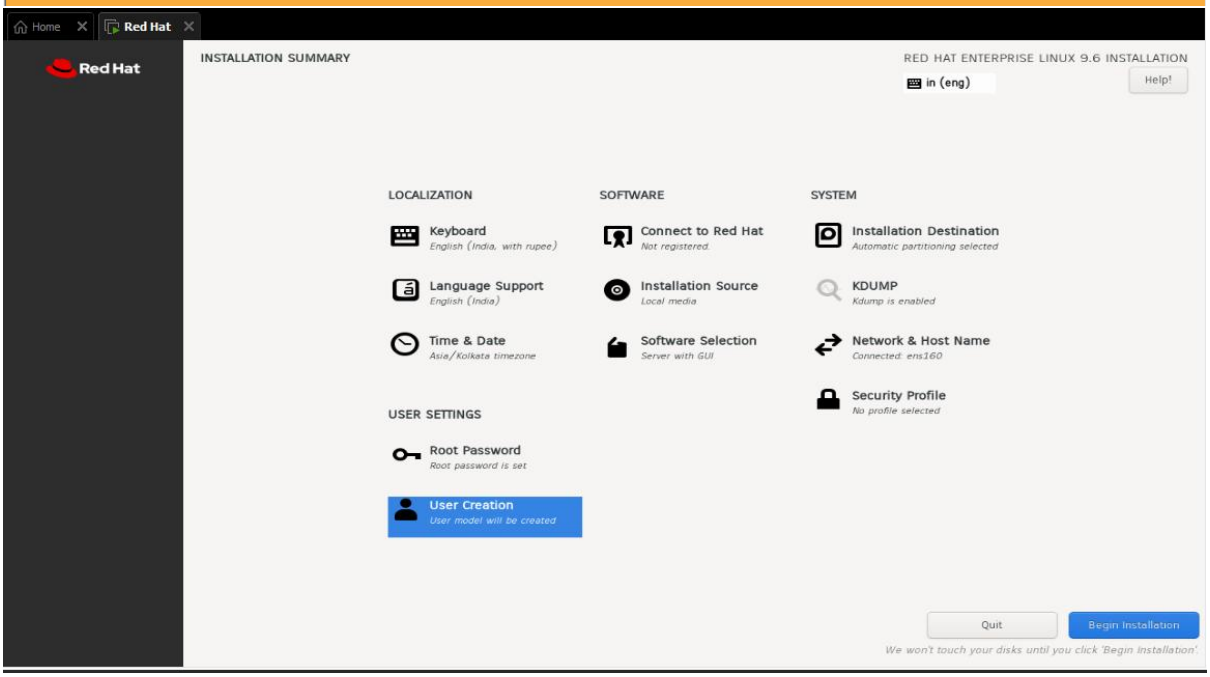

Home
Red Hat
Red Hat
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
us
Help!
WELCOME TO RED HAT ENTERPRISE LINUX 9.6.
What language would you like to use during the installation process?
हिन्दी Hindi
العربية Arabic
English English
Français French
Deutsch German
日本語 Japanese
中文 Mandarin Chinese
Русский Russian
Español Spanish
Afrikaans Afrikaans
አማርኛ Amharic
অসমীয়া Assamese
Asturianu Asturian
Беларуская Belarusian
Български Bulgarian
Type here to search.
English (United States)
English (United Kingdom)
English (India)
English (Australia)
English (Canada)
English (Denmark)
English (Ireland)
English (New Zealand)
English (Nigeria)
English (Hong Kong SAR China)
English (Philippines)
English (Singapore)
English (South Africa)
English (Zambia)
English (Zimbabwe)
English (Botswana)
English (Antigua & Barbuda)
Quit
Continue

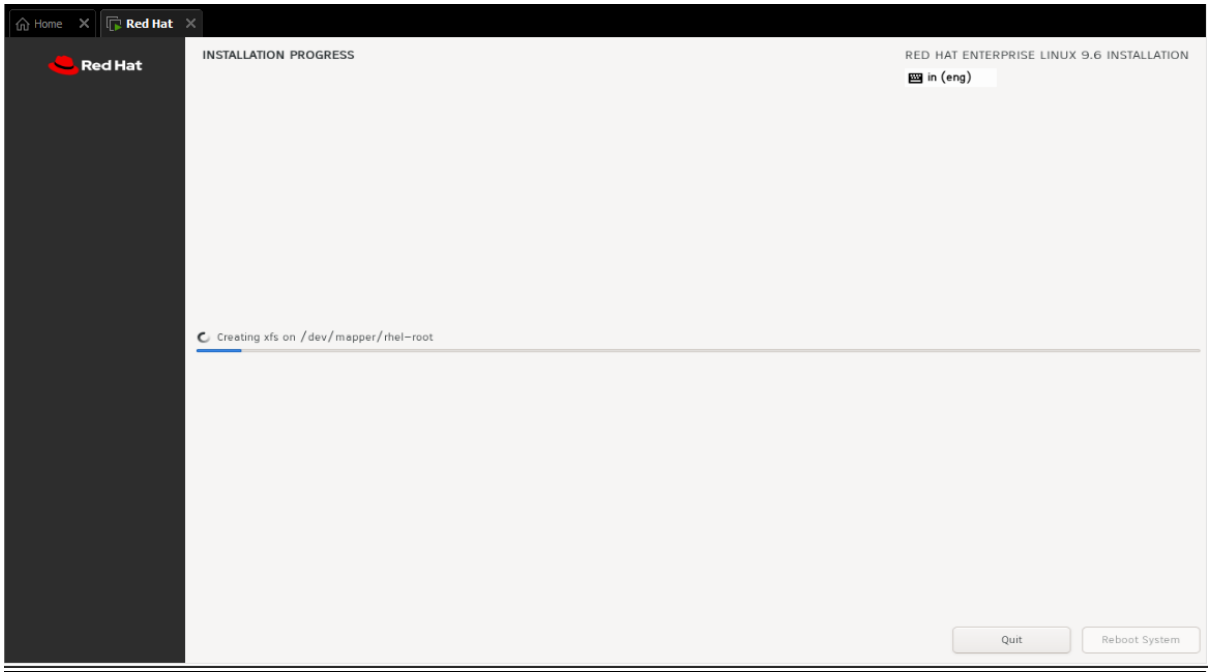

Home
Red Hat
Red Hat
INSTALLATION SUMMARY
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
LOCALIZATION
Keyboard
English (India, with rupee)
Language Support
English (India)
Time & Date
Asia/Kolkata timezone
SOFTWARE
Connect to Red Hat
Not registered.
Installation Source
Local media
Software Selection
Server with GUI
SYSTEM
Installation Destination
Automatic partitioning selected
KDUMP
Kdump is enabled
Network & Host Name
Connected: ens160
Security Profile
No profile selected
USER SETTINGS
Root Password
Root account is disabled
User Creation
No user will be created
Quit
Begin Installation
We won't touch your disks until you click 'Begin Installation'.
Please complete items marked with this icon before continuing to the next step.

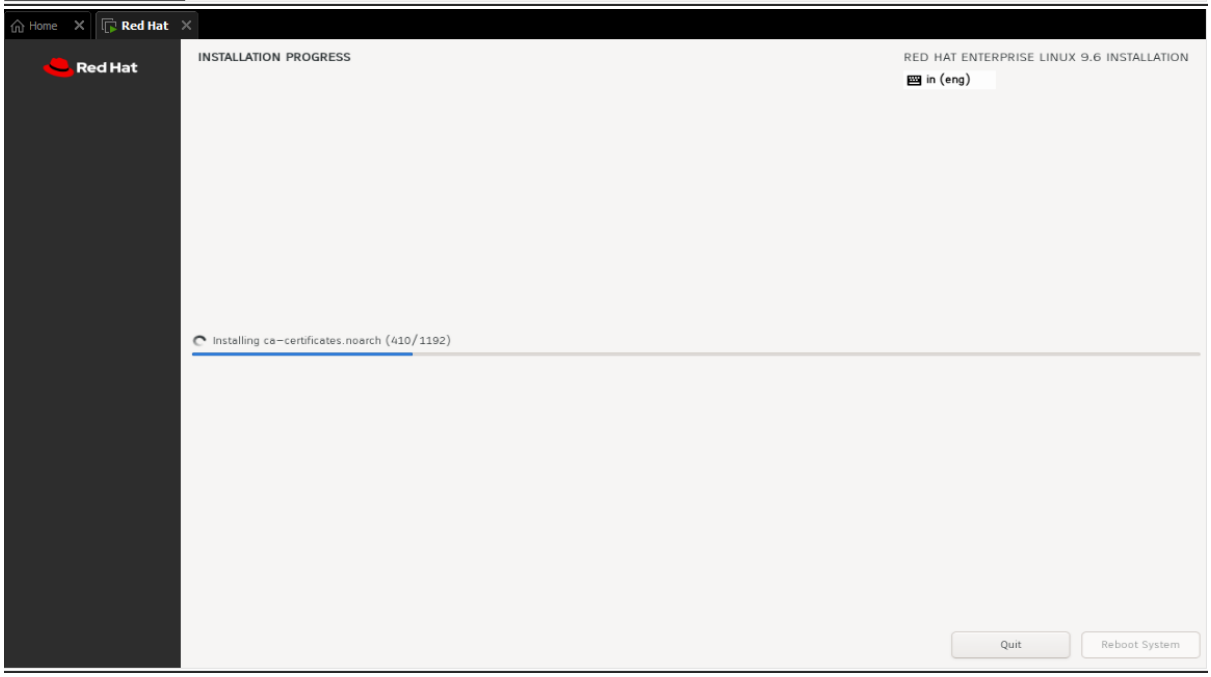

Home
Red Hat
INSTALLATION DESTINATION
Done
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
Device Selection
Select the device(s) you'd like to install to. They will be left untouched until you click on the main menu's "Begin Installation" button.
Local Standard Disks
20 GiB
VMware Virtual NVMe Disk
nvme0n1 / 20 GiB free
Disks left unselected here will not be touched.
Specialized & Network Disks
Add a disk...
Disks left unselected here will not be touched.
Storage Configuration
Automatic
Custom
I would like to make additional space available.
Encryption
Encrypt my data. You'll set a passphrase next.
Full disk summary and boot loader...
1 disk selected; 20 GiB capacity; 20 GiB free
Refresh...


Home
Red Hat
ROOT PASSWORD
Done
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
The root account is used for administering the system. Enter a password for the root user.
Root Password:
Too short
Confirm:
Lock root account
Allow root SSH login with password
The password is too short. You will have to press Done twice to confirm it.

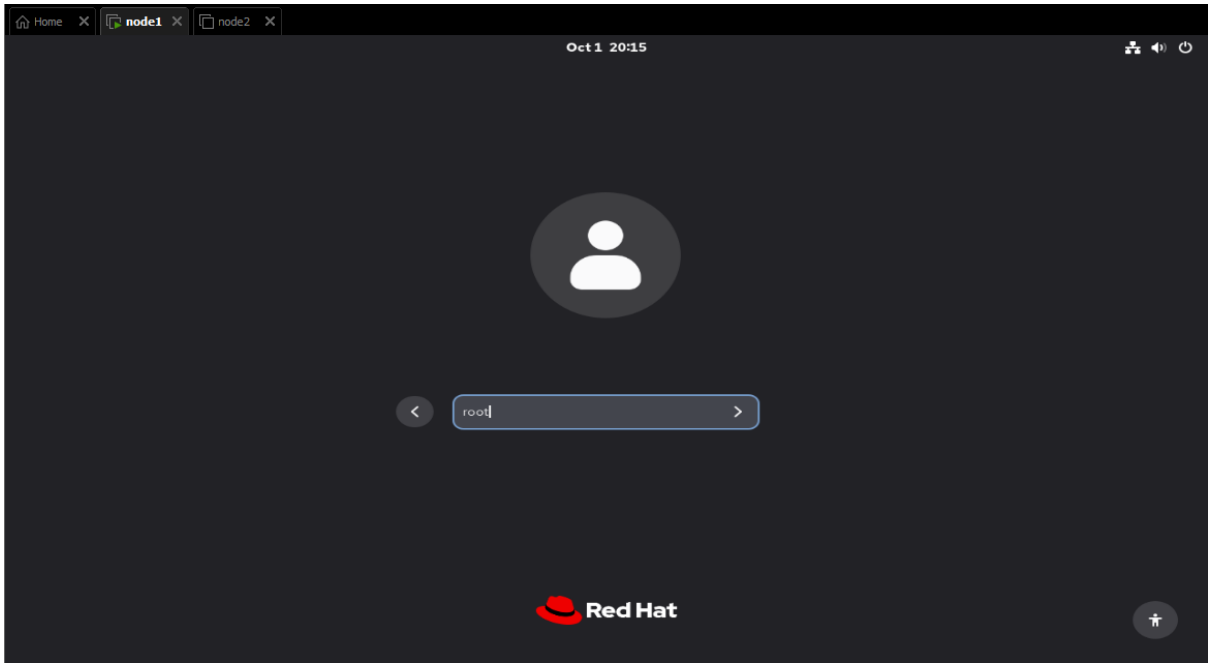

Home
Red Hat
CREATE USER
Done
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
Full name
model
User name
model
Make this user administrator
Require a password to use this account
Password
Too short
Confirm password
Advanced...
The password is too short. You will have to press Done twice to confirm it.

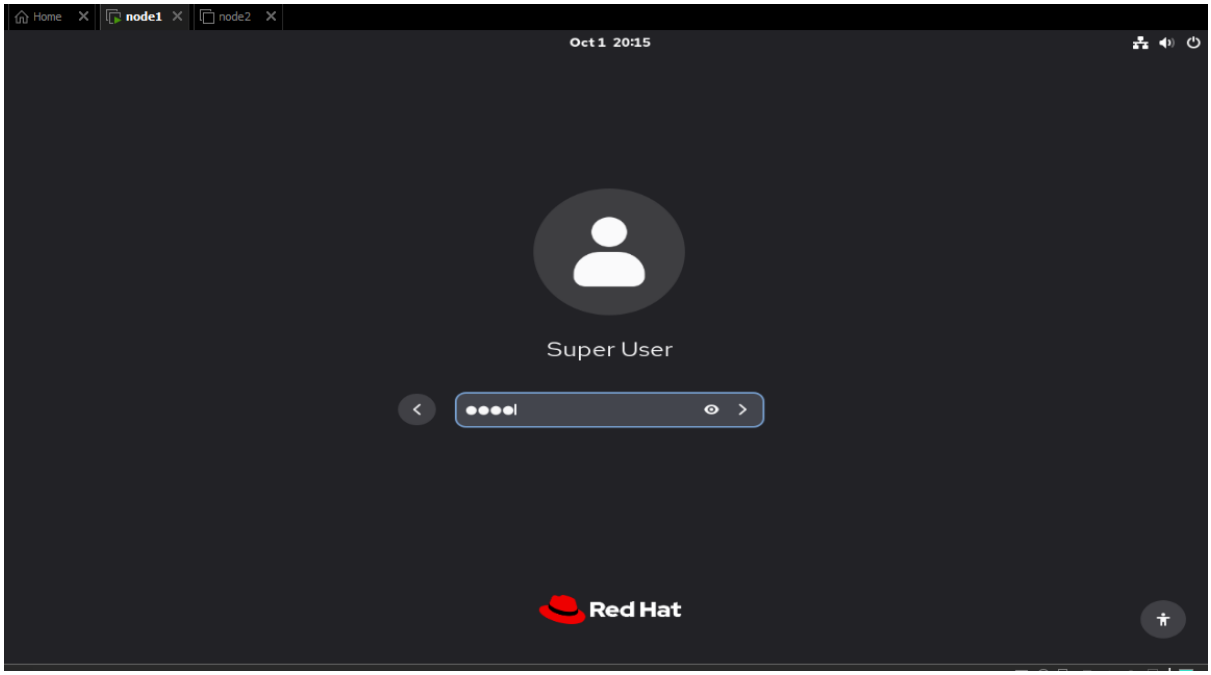

Home
Red Hat
Red Hat
INSTALLATION SUMMARY
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Help!
LOCALIZATION
Keyboard
English (India, with rupee)
Language Support
English (India)
Time & Date
Asia/Kolkata timezone
SOFTWARE
Connect to Red Hat
Not registered.
Installation Source
Local media
Software Selection
Server with GUI
SYSTEM
Installation Destination
Automatic partitioning selected
KDUMP
Kdump is enabled
Network & Host Name
Connected: ens160
Security Profile
No profile selected
USER SETTINGS
Root Password
Root password is set
User Creation
User model will be created
Quit
Begin Installation
We won't touch your disks until you click 'Begin Installation'.


Home
Red Hat
Red Hat
INSTALLATION PROGRESS
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Creating xfs on /dev/mapper/rhel-root
Quit
Reboot System


Home
Red Hat
Red Hat
INSTALLATION PROGRESS
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Installing ca-certificates.noarch (410/1192)
Quit
Reboot System


Home
Red Hat
Red Hat
INSTALLATION PROGRESS
RED HAT ENTERPRISE LINUX 9.6 INSTALLATION
in (eng)
Complete!
Red Hat Enterprise Linux is now successfully installed and ready for you to use!
Go ahead and reboot your system to start using it!
Reboot System
Use of this product is subject to the license agreement found at /usr/share/redhat-release/EULA


Home
node1
node2
Oct 1 20:15
root
Red Hat


Home
node1
node2
Oct 1 20:15
Super User
Red Hat



Successfully logged-in RHEL desktop.

B. Linux Graphical Environment

1. Display/Login Manager

Check the display manager:

systemctl status display-manager

Expected Output

● gdm.service - GNOME Display Manager

Loaded: loaded

Active: active (running)

Run:

systemctl status display-manager

Print this terminal output.



---

2. Desktop Environment

RHEL Workstation commonly uses GNOME.

Execute:
echo $XDG_CURRENT_DESKTOP
echo $DESKTOP_SESSION
Expected Output
GNOME
gnome
Check operating system:
cat /etc/os-release
Expected Output
NAME="Red Hat Enterprise Linux"
VERSION="10..."
ID="rhel"



C. Linux Directory Structure
Execute:
ls /
Typical directories include:
bin
boot
dev
etc
home
lib
lib64
media
mnt
opt
proc
root
run

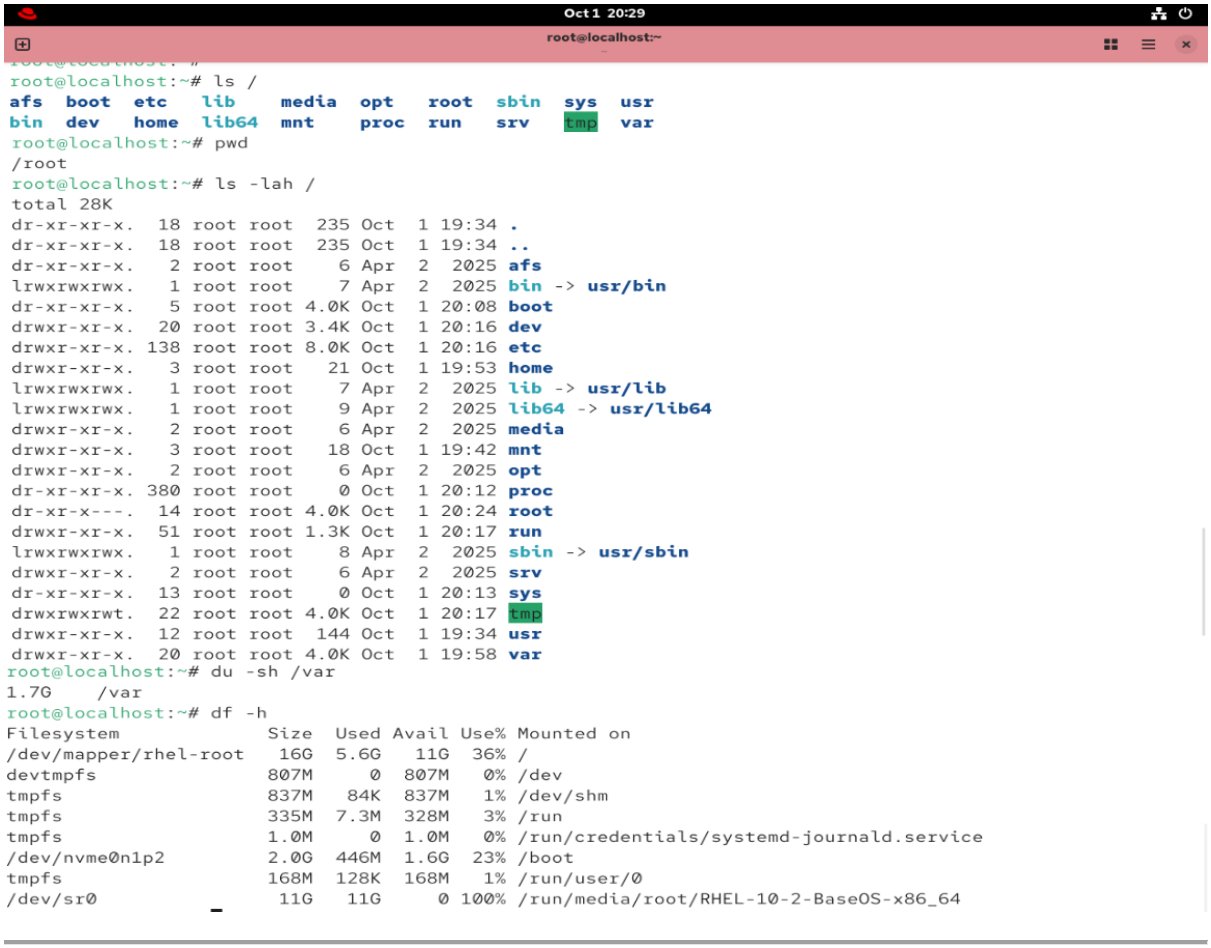
sbin
srv
sys
tmp
usr
var

Important Directories

Directory	Purpose
/	Root directory
/boot	Boot-related files
/etc	System configuration
/home	User home directories
/root	Root user's home
/tmp	Temporary files
/usr	User applications and system resources
/var	Variable data/logs
/dev	Device files
/proc	Process/kernel information
/sys	Kernel/device information
/mnt	Temporary mount point
/opt	Optional software

Commands

pwd
ls -lah /
du -sh /var
df -h


Oct 1 20:29
root@localhost:~
root@localhost:~# ls /
afs boot etc lib media opt root sbin sys usr
bin dev home lib64 mnt proc run srv tmp var
root@localhost:~# pwd
/root
root@localhost:~# ls -lah /
total 28K
dr-xr-xr-x. 18 root root 235 Oct 1 19:34 .
dr-xr-xr-x. 18 root root 235 Oct 1 19:34 ..
dr-xr-xr-x. 2 root root 6 Apr 2 2025 afs
lrwxrwxrwx. 1 root root 7 Apr 2 2025 bin -> usr/bin
dr-xr-xr-x. 5 root root 4.0K Oct 1 20:08 boot
drwxr-xr-x. 20 root root 3.4K Oct 1 20:16 dev
drwxr-xr-x. 138 root root 8.0K Oct 1 20:16 etc
drwxr-xr-x. 3 root root 21 Oct 1 19:53 home
lrwxrwxrwx. 1 root root 7 Apr 2 2025 lib -> usr/lib
lrwxrwxrwx. 1 root root 9 Apr 2 2025 lib64 -> usr/lib64
drwxr-xr-x. 2 root root 6 Apr 2 2025 media
drwxr-xr-x. 3 root root 18 Oct 1 19:42 mnt
drwxr-xr-x. 2 root root 6 Apr 2 2025 opt
dr-xr-xr-x. 380 root root 0 Oct 1 20:12 proc
dr-xr-x---. 14 root root 4.0K Oct 1 20:24 root
drwxr-xr-x. 51 root root 1.3K Oct 1 20:17 run
lrwxrwxrwx. 1 root root 8 Apr 2 2025 sbin -> usr/sbin
drwxr-xr-x. 2 root root 6 Apr 2 2025 srv
dr-xr-xr-x. 13 root root 0 Oct 1 20:13 sys
drwxrwxrwt. 22 root root 4.0K Oct 1 20:17 tmp
drwxr-xr-x. 12 root root 144 Oct 1 19:34 usr
drwxr-xr-x. 20 root root 4.0K Oct 1 19:58 var
root@localhost:~# du -sh /var
1.7G /var
root@localhost:~# df -h
Filesystem Size Used Avail Use% Mounted on
/dev/mapper/rhel-root 16G 5.6G 11G 36% /
devtmpfs 807M 0 807M 0% /dev
tmpfs 837M 84K 837M 1% /dev/shm
tmpfs 335M 7.3M 328M 3% /run
tmpfs 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2 2.0G 446M 1.6G 23% /boot
tmpfs 168M 128K 168M 1% /run/user/0
/dev/sr0 11G 11G 0 100% /run/media/root/RHEL-10-2-BaseOS-x86_64

PRACTICAL 2
Basic System Commands and Input/Output Redirection

Aim

To study basic Linux commands and demonstrate standard input, standard output, error redirection and pipes.

Commands

Display current directory

pwd

List files

ls
ls -l
ls -la

Display date

date

Display current user

whoami

Display logged-in users

who

System information

uname -a

Hostname

hostname

Input/Output Redirection

1. >

Creates/overwrites a file.

echo "Linux Practical" > practical.txt
cat practical.txt
Output:
Linux Practical

---

2. >>

Appends data.

echo "TYIT RHEL Practical" >> practical.txt
cat practical.txt
Output:
Linux Practical
TYIT RHEL Practical

---

3. Pipe |

ls -l | less
Another example:
ps aux | grep ssh

---

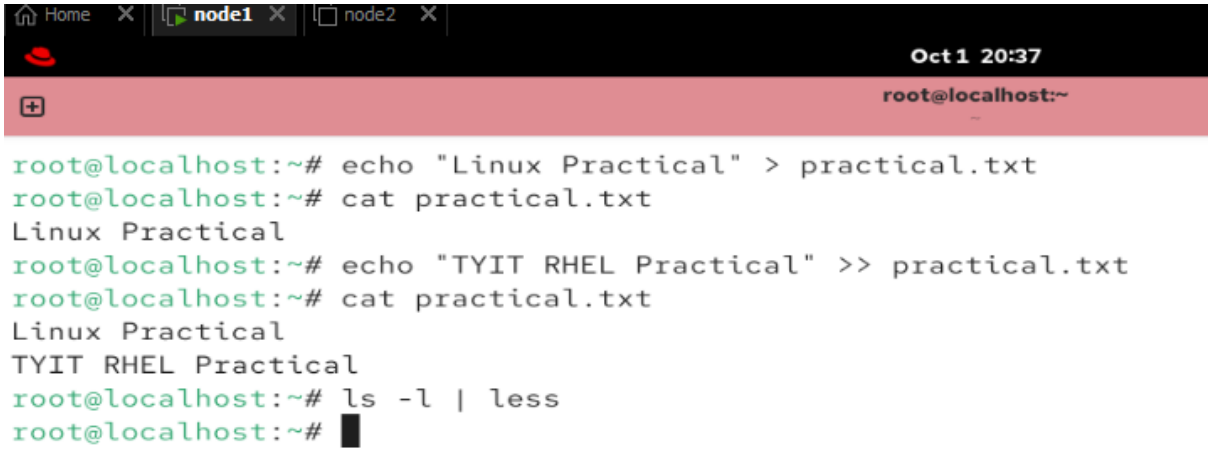
4. Error Redirection 2>

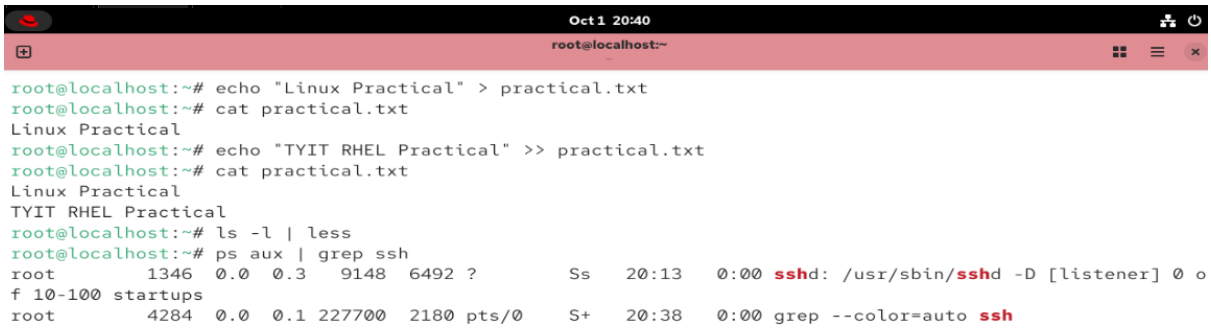
ls /wrong-directory 2> error.txt
cat error.txt

---

5. Redirect output and errors
command > output.txt 2> error.txt

Students should execute:
echo "Linux" > a.txt
echo "RHEL" >> a.txt
cat a.txt
ls /wrong 2> error.txt
cat error.txt





PRACTICAL 3

grep, Regular Expressions and Text Processing

Aim

To analyse text using grep, regular expressions and text-processing commands.

Create a file:

cat > students.txt

Enter:

Amit TYIT 85
Neha TYIT 92
Rahul SYIT 76
Priya TYIT 88
Rohan SYIT 69
Sneha TYIT 95

Press **Ctrl+D**.

---

grep

grep TYIT students.txt

Expected:

Amit TYIT 85
Neha TYIT 92
Priya TYIT 88
Sneha TYIT 95

Case-insensitive grep

grep -i tyit students.txt

Line numbers

grep -n TYIT students.txt

Regular Expression

Find lines beginning with A:

grep '^A' students.txt

Find lines ending with 5:

grep '5$' students.txt

Find exactly three digits:

grep -E '[0-9]{3}' students.txt

Text File Commands

cat

cat students.txt

head

Your syllabus says hesd; the correct command is:

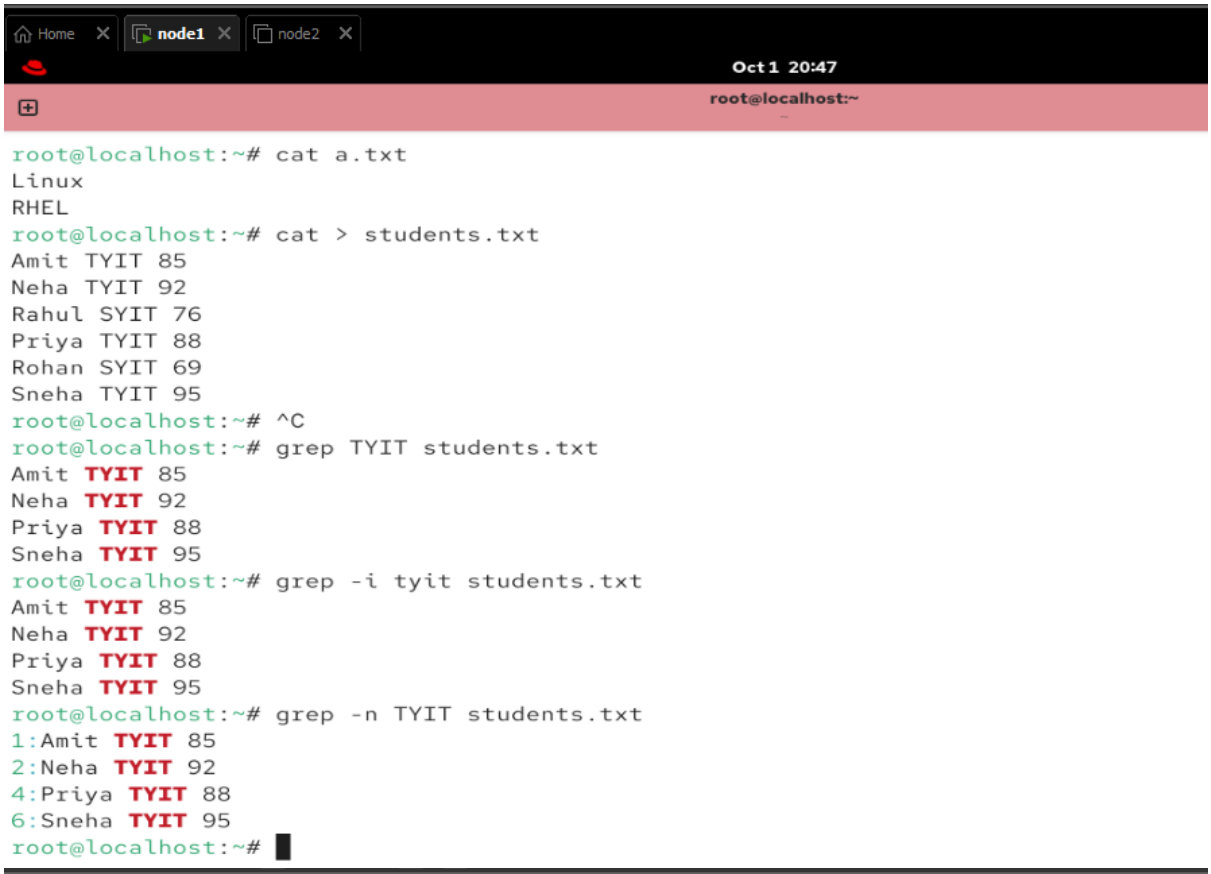
head students.txt

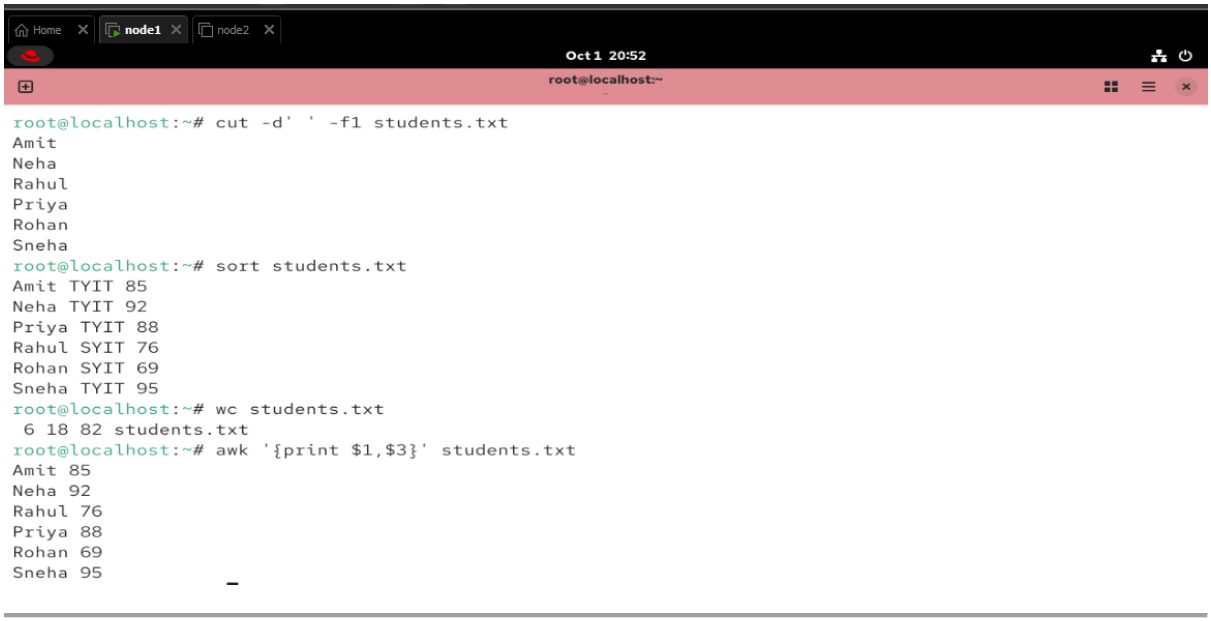
tail

tail students.txt

less

less students.txt
Press q to exit.
cut
cut -d' ' -f1 students.txt
sort
sort students.txt
wc
wc students.txt
awk
awk '{print $1,$3}' students.txt
Expected:
Amit 85
Neha 92
Rahul 76
Priya 88
Rohan 69
Sneha 95


Home
node1
node2
Oct 1 20:47
root@localhost:~
root@localhost:~# cat a.txt
Linux
RHEL
root@localhost:~# cat > students.txt
Amit TYIT 85
Neha TYIT 92
Rahul SYIT 76
Priya TYIT 88
Rohan SYIT 69
Sneha TYIT 95
root@localhost:~# ^C
root@localhost:~# grep TYIT students.txt
Amit TYIT 85
Neha TYIT 92
Priya TYIT 88
Sneha TYIT 95
root@localhost:~# grep -i tyit students.txt
Amit TYIT 85
Neha TYIT 92
Priya TYIT 88
Sneha TYIT 95
root@localhost:~# grep -n TYIT students.txt
1:Amit TYIT 85
2:Neha TYIT 92
4:Priya TYIT 88
6:Sneha TYIT 95
root@localhost:~#


Home
node1
node2
Oct 1 20:52
root@localhost:~
root@localhost:~# cut -d' ' -f1 students.txt
Amit
Neha
Rahul
Priya
Rohan
Sneha
root@localhost:~# sort students.txt
Amit TYIT 85
Neha TYIT 92
Priya TYIT 88
Rahul SYIT 76
Rohan SYIT 69
Sneha TYIT 95
root@localhost:~# wc students.txt
6 18 82 students.txt
root@localhost:~# awk '{print $1,$3}' students.txt
Amit 85
Neha 92
Rahul 76
Priya 88
Rohan 69
Sneha 95

PRACTICAL 4
File and Directory Management

Aim

To create, delete, copy and move files and directories.

Create directory

mkdir tyit_lab

cd tyit_lab

Create files

touch file1.txt file2.txt file3.txt

List files

ls -l

Copy file

cp file1.txt file1_backup.txt

Move/rename

mv file2.txt renamed.txt

Create directory

mkdir documents

Copy directory

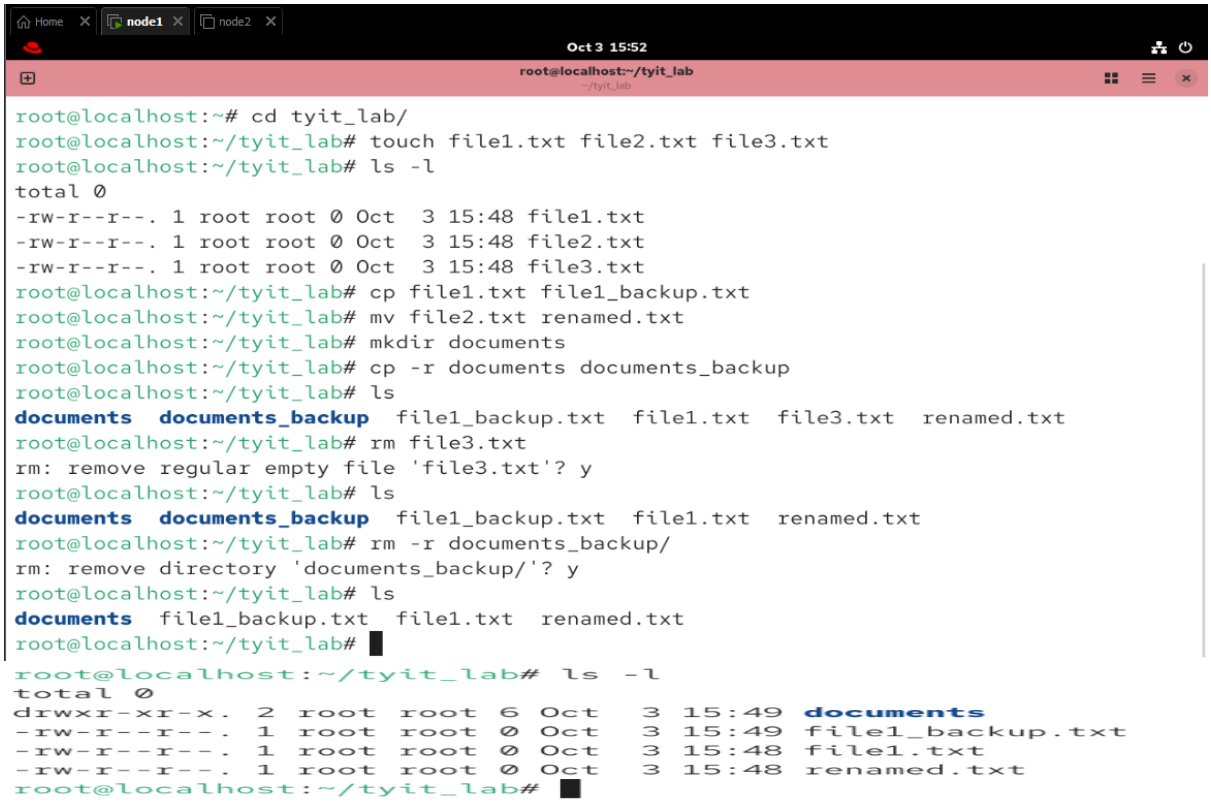
cp -r documents documents_backup

Remove file

rm file3.txt

Remove directory

rm -r documents_backup



PRACTICAL 5

Links, Archives and Compression

Aim

To create hard/soft links and manage compressed and archived files.

A. Hard Link

Create a file:

echo "TYIT Linux" > original.txt

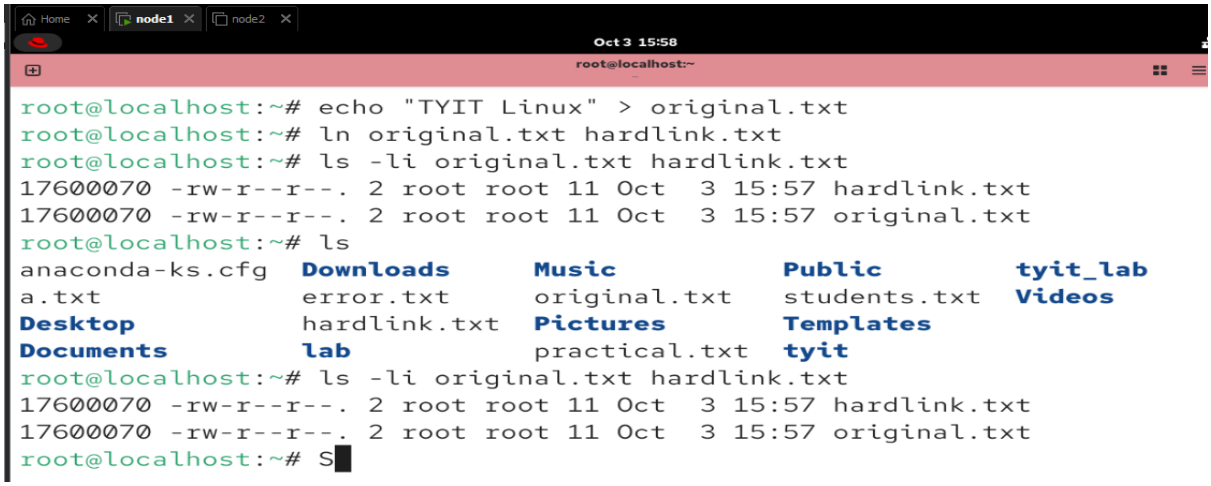
Create hard link:

ln original.txt hardlink.txt

Check inode:

ls -li original.txt hardlink.txt

Both files should have the same inode number.



B. Soft Link

ln -s original.txt softlink.txt

Check:

ls -l softlink.txt

Expected:

softlink.txt -> original.txt

```
root@localhost:~# ln -s original.txt softlink.txt
root@localhost:~# ls -l softlink.txt
lrwxrwxrwx. 1 root root 12 Oct  3 15:59 softlink.txt -> original.txt
root@localhost:~#
```

C. tar

Create archive:

tar -cvf tyit.tar original.txt hardlink.txt

List archive:

tar -tvf tyit.tar

Extract:

mkdir extracted

tar -xvf tyit.tar -C extracted

D. gzip

gzip original.txt

Check:

ls

Uncompress:

gunzip original.txt.gz

E. bzip2

bzip2 original.txt

Uncompress:

bunzip2 original.txt.bz2

F. star

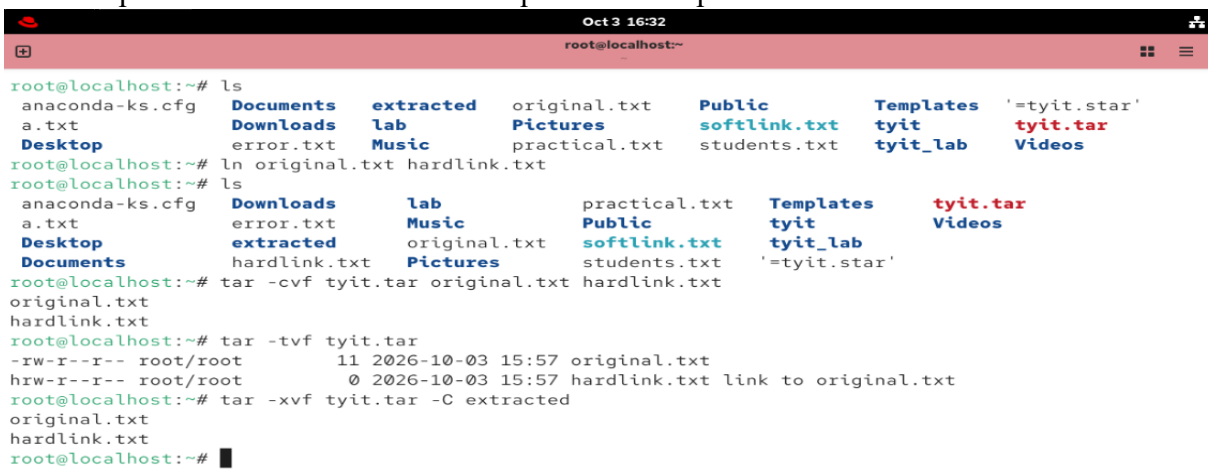
If star is installed:

star -c -f=tyit.star original.txt

List:

star -t -f=tyit.star

If star is not installed in your RHEL installation, install the package through the configured RHEL repositories or demonstrate the equivalent tar operation.



```
root@localhost:~# tar -c -f=tyit.star original.txt
root@localhost:~# star -t -f=tyit.star
bash: star: command not found...
Similar command is: 'tar'
root@localhost:~# tar -t -f=tyit.star
original.txt
root@localhost:~#
```

```
root@localhost:~# gzip original.txt
root@localhost:~# ls
anaconda-ks.cfg  Downloads  Music             Public        tyit
a.txt            error.txt  original.txt.gz   softlink.txt  tyit_lab
Desktop          extracted  Pictures          students.txt  tyit.tar
Documents        lab        practical.txt     Templates     Videos
root@localhost:~# gunzip original.txt.gz
root@localhost:~# ls
anaconda-ks.cfg  Downloads  Music             Public        tyit
a.txt            error.txt  original.txt      softlink.txt  tyit_lab
Desktop          extracted  Pictures          students.txt  tyit.tar
Documents        lab        practical.txt     Templates     Videos
root@localhost:~#
```

PRACTICAL 6
File Ownership, Permissions and ACL

Aim
To manage Linux ownership, standard permissions, special permissions and ACLs.

A. View permissions
ls -l
Example:
-rw-r--r-- 1 student student 15 Sep 20 10:20 file.txt
Permission format:

```
- rw- r-- r--
  |  |  |
  |  |  +--- Others
  |  +------- Group
  +---------- Owner
```

B. chmod
Create file:
touch permission.txt
Set permissions:
chmod 755 permission.txt
Check:
ls -l permission.txt

Numeric Permissions

Number	Permission
0	---
1	--x
2	-w-
3	-wx
4	r--
5	r-x
6	rw-
7	rwx

Example:
chmod 644 file.txt

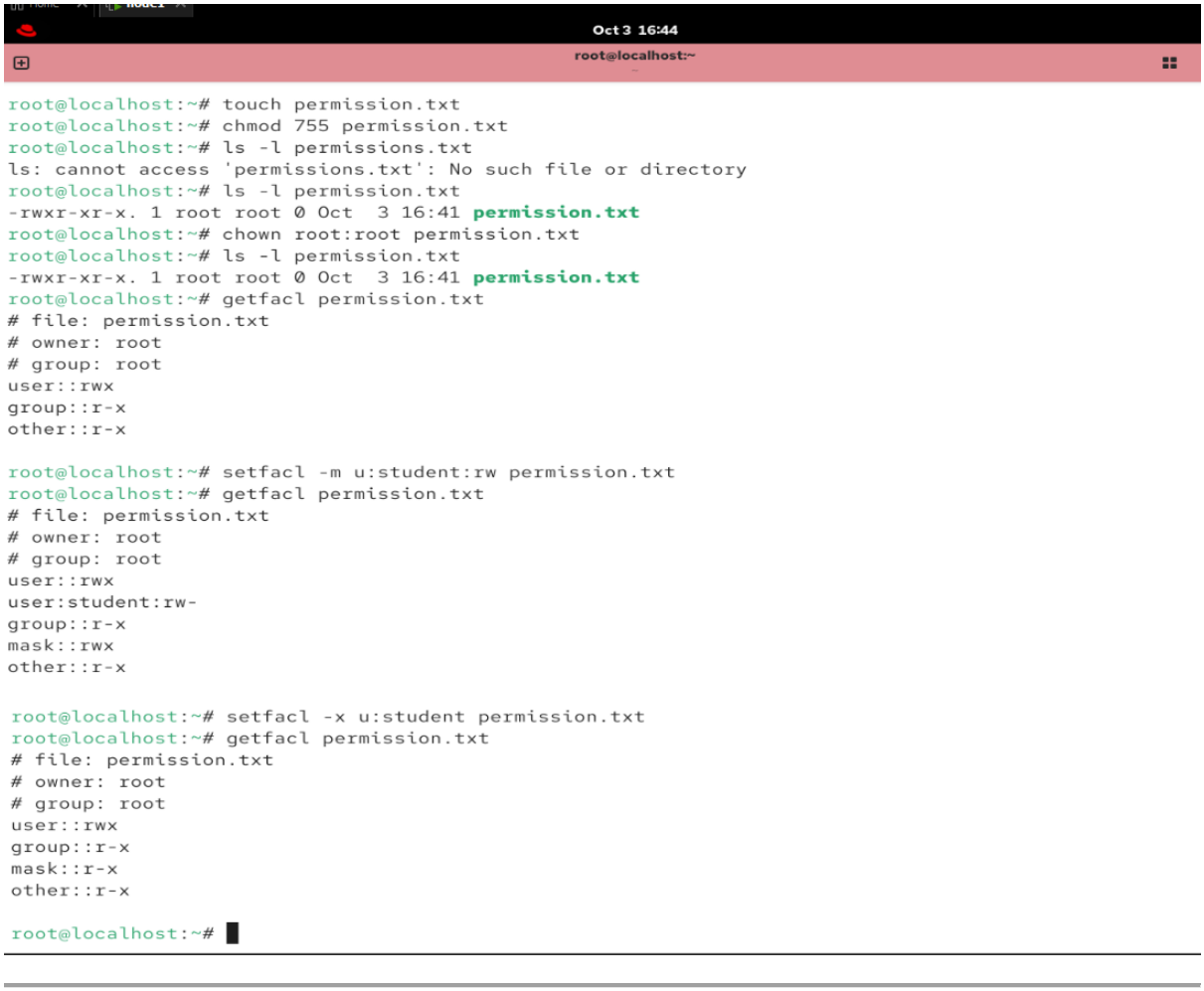
C. chown
sudo chown root:root permission.txt
Check:
ls -l permission.txt

D. umask
umask
Temporarily set:

umask 027
Create:
touch umaskfile
Check:
ls -l umaskfile

E. set-GID Directory
Create group:
sudo groupadd project
Create directory:
sudo mkdir /project
Change group:
sudo chgrp project /project
Set set-GID:
sudo chmod 2775 /project
Verify:
ls -ld /project
The group directory should show s in the group permission position.

F. ACL
Check ACL:
getfacl permission.txt
Set ACL:
sudo setfacl -m u:student:rw permission.txt
Check:
getfacl permission.txt
Remove ACL:
sudo setfacl -x u:student permission.txt


Oct 3 16:44
root@localhost:~
root@localhost:~# touch permission.txt
root@localhost:~# chmod 755 permission.txt
root@localhost:~# ls -l permissions.txt
ls: cannot access 'permissions.txt': No such file or directory
root@localhost:~# ls -l permission.txt
-rwxr-xr-x. 1 root root 0 Oct 3 16:41 permission.txt
root@localhost:~# chown root:root permission.txt
root@localhost:~# ls -l permission.txt
-rwxr-xr-x. 1 root root 0 Oct 3 16:41 permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
group::r-x
other::r-x
root@localhost:~# setfacl -m u:student:rw permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
user:student:rw-
group::r-x
mask::rwx
other::r-x
root@localhost:~# setfacl -x u:student permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
group::r-x
mask::r-x
other::r-x
root@localhost:~#

PRACTICAL 7
Users, Groups and Superuser Administration

Aim

To create and manage users/groups, configure passwords, password aging and superuser access.

Create User

sudo useradd labuser
Set password:
sudo passwd labuser

Create Group

sudo groupadd tyitgroup
Add user to group:
sudo usermod -aG tyitgroup labuser
Verify:
id labuser

Password Aging

sudo chage -l labuser
Set maximum password age:
sudo chage -M 90 labuser
Set minimum age:
sudo chage -m 1 labuser
Set warning period:
sudo chage -W 7 labuser
Verify:
sudo chage -l labuser

---

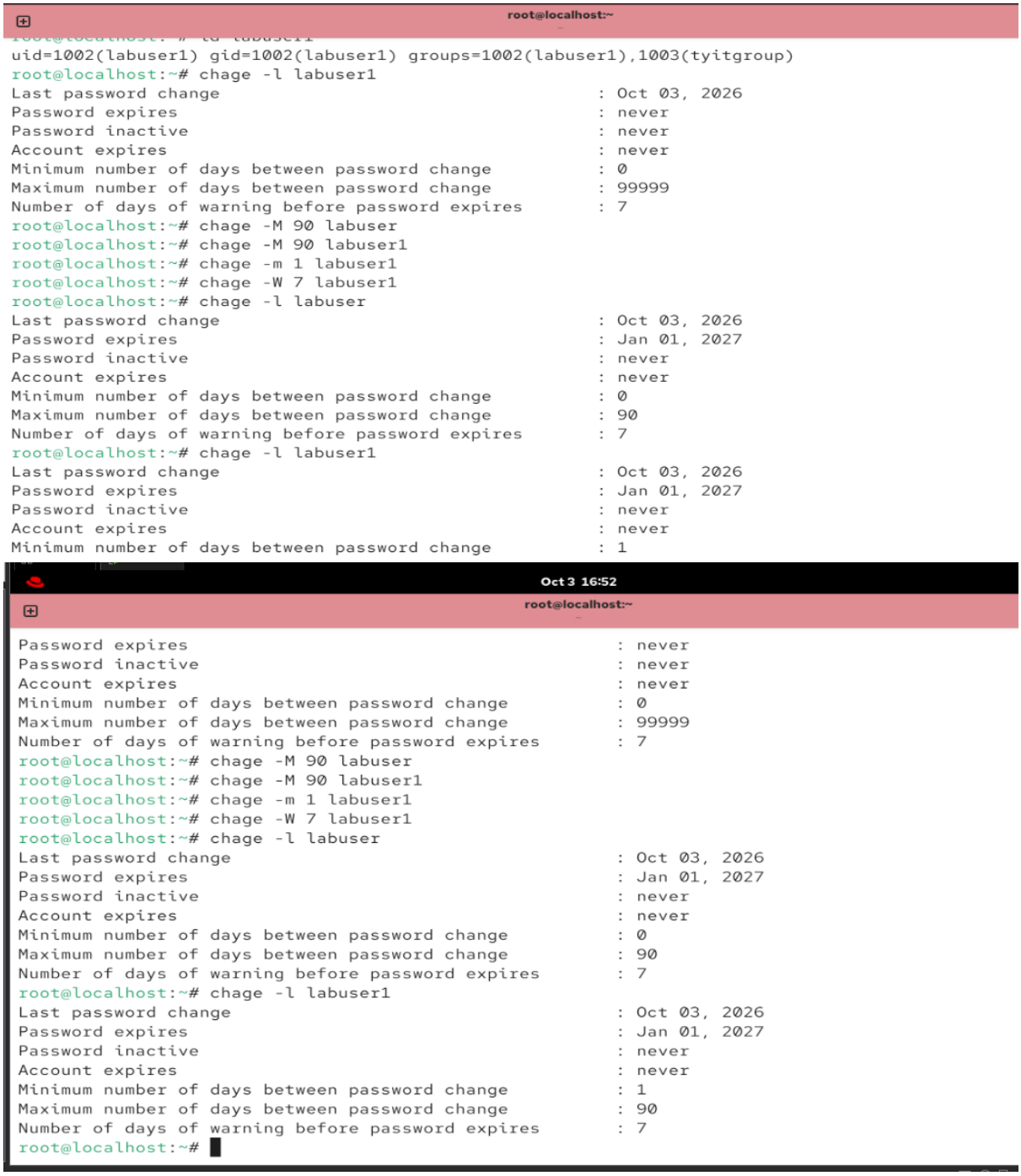
su

su - labuser
Return:
exit

---

sudo

Check:
sudo -l
Run administrative command:
sudo whoami
Expected:
root


root@localhost:~
uid=1002(labuser1) gid=1002(labuser1) groups=1002(labuser1),1003(tyitgroup)
root@localhost:~# chage -l labuser1
Last password change : Oct 03, 2026
Password expires : never
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
root@localhost:~# chage -M 90 labuser
root@localhost:~# chage -M 90 labuser1
root@localhost:~# chage -m 1 labuser1
root@localhost:~# chage -W 7 labuser1
root@localhost:~# chage -l labuser
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
root@localhost:~# chage -l labuser1
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 1
Oct 3 16:52
root@localhost:~
Password expires : never
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
root@localhost:~# chage -M 90 labuser
root@localhost:~# chage -M 90 labuser1
root@localhost:~# chage -m 1 labuser1
root@localhost:~# chage -W 7 labuser1
root@localhost:~# chage -l labuser
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
root@localhost:~# chage -l labuser1
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 1
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
root@localhost:~#


root@localhost:~
Number of days of warning before password expires : 7
root@localhost:~# chage -l labuser1
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 1
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
root@localhost:~# su - labuser1
labuser1@localhost:~$ exit
logout
root@localhost:~# -l
bash: -l: command not found...
root@localhost:~# sudo -l
Matching Defaults entries for root on localhost:
!visiblepw, always_set_home, match_group_by_gid, always_query_group_plugin, env_reset,
env_keep="COLORS DISPLAY HOSTNAME HISTSIZE KDEDIR LS_COLORS", env_keep+="MAIL PS1 PS2 QTDIR USERNAME
LANG LC_ADDRESS LC_CTYPE", env_keep+="LC_COLLATE LC_IDENTIFICATION LC_MEASUREMENT LC_MESSAGES",
env_keep+="LC_MONETARY LC_NAME LC_NUMERIC LC_PAPER LC_TELEPHONE", env_keep+="LC_TIME LC_ALL LANGUAGE
LINGUAS _XKB_CHARSET XAUTHORITY", secure_path=/sbin\:/bin\:/usr/sbin\:/usr/bin
User root may run the following commands on localhost:
(ALL) ALL
root@localhost:~# whoami
root
root@localhost:~#

PRACTICAL 8
Software Package Management

Aim
To manage software using DNF/YUM and RPM.
Current RHEL 10 documentation identifies **DNF** as the package-management tool.

A. DNF
Check repositories:
sudo dnf repolist
Search package:
dnf search httpd
Install:
sudo dnf install httpd
Check package:
dnf info httpd
Update:
sudo dnf update
Remove:
sudo dnf remove httpd
RHEL documentation confirms commands such as dnf search, dnf repolist, dnf install, dnf remove, and dnf update.

B. YUM
On modern RHEL systems:
yum repolist
yum search httpd
sudo yum install httpd
yum may act as a compatibility interface to DNF on current RHEL systems.

C. RPM
Query:
rpm -q bash
Package information:
rpm -qi bash
List files:
rpm -ql bash
Verify:
rpm -V bash

D. Local RPM
For a local RPM:
sudo dnf install ./package-name.rpm
RHEL 10 documentation explicitly supports installing a local RPM using dnf install <path_to_RPM_file>.

PRACTICAL 9
Process Management

Aim

To monitor, analyse and control Linux processes.

View processes

ps

ps aux

ps -ef

Find a process

ps aux | grep ssh

Process tree

pstree

If unavailable:

ps --forest

---

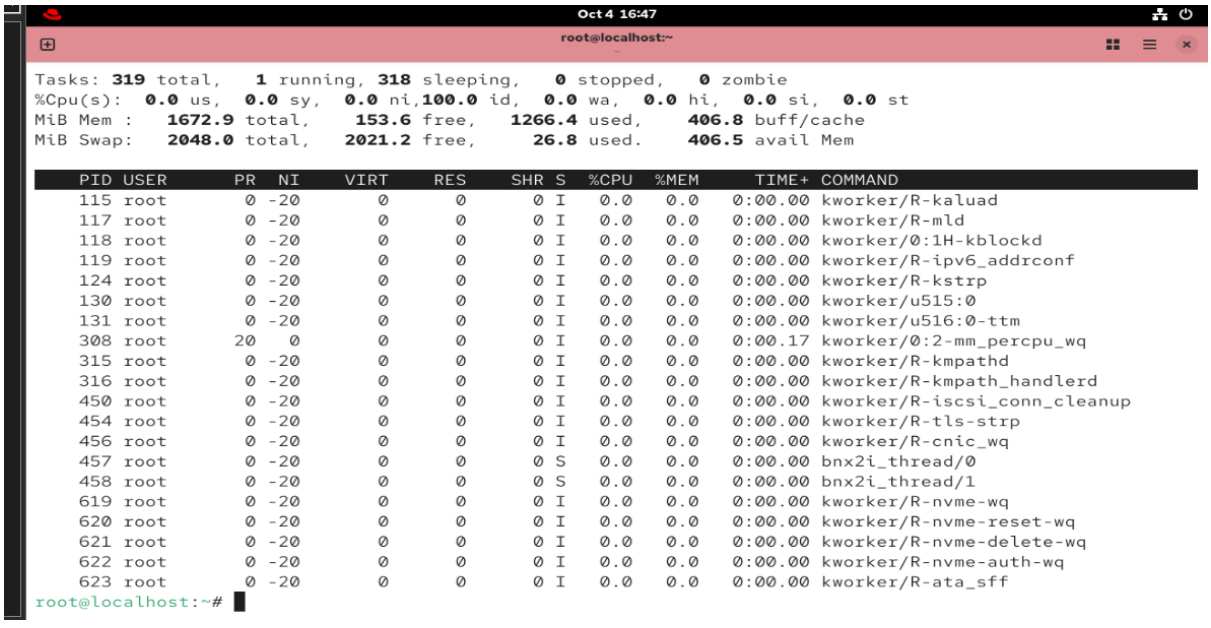
top

top

Press:

q

to exit.



Identify resource-intensive processes

ps -eo pid,user,%cpu,%mem,comm --sort=-%cpu | head

Memory:

ps -eo pid,user,%cpu,%mem,comm --sort=-%mem | head

Kill Process

Find PID:
pgrep sleep
Terminate:
kill PID
If necessary:
kill -9 PID
Use kill -9 only when normal termination does not work.

---

tuned
Check:
systemctl status tuned
List profiles:
tuned-adm list
Current profile:
tuned-adm active

PRACTICAL 10
Shell Job Management

Aim

To manage foreground and background jobs.

Run:

sleep 300

Press:

Ctrl+Z

Resume in background:

bg

Display jobs:

jobs

Bring to foreground:

fg

Run directly in background:

sleep 300 &

Check:

jobs

Terminate:

kill %1



PRACTICAL 11
Storage Management

Aim

To study MBR/GPT, partitions, file systems, mounting, UUID, labels, LVM and swap.

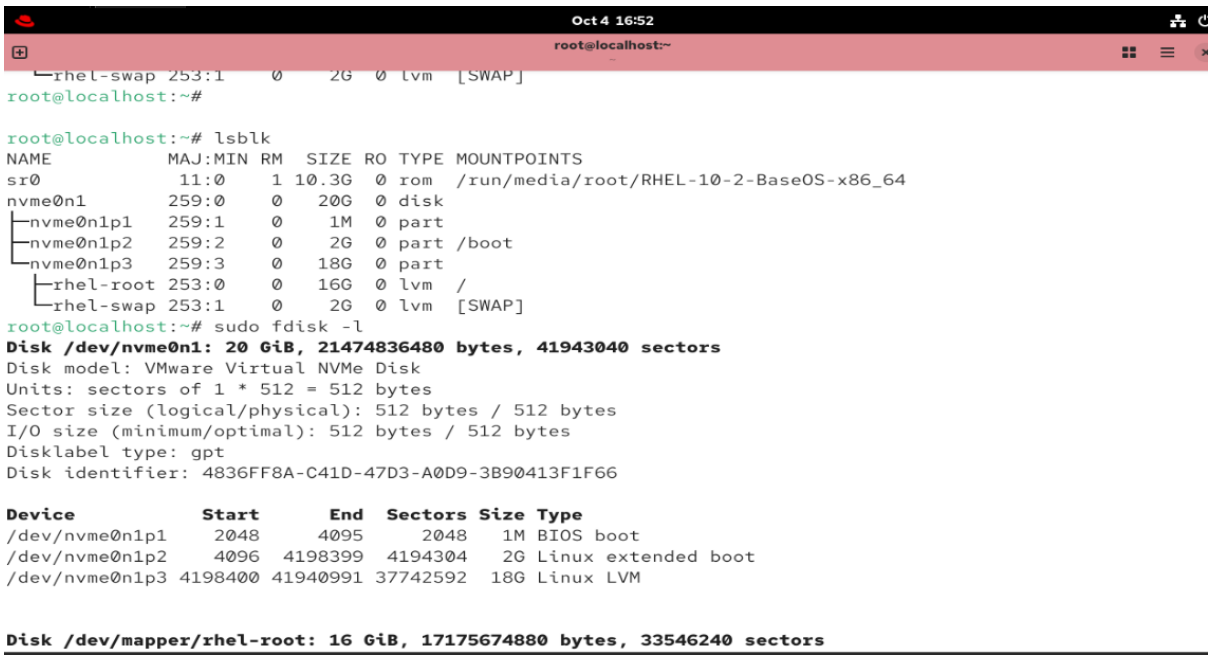
WARNING: Perform this practical only on an additional virtual disk.

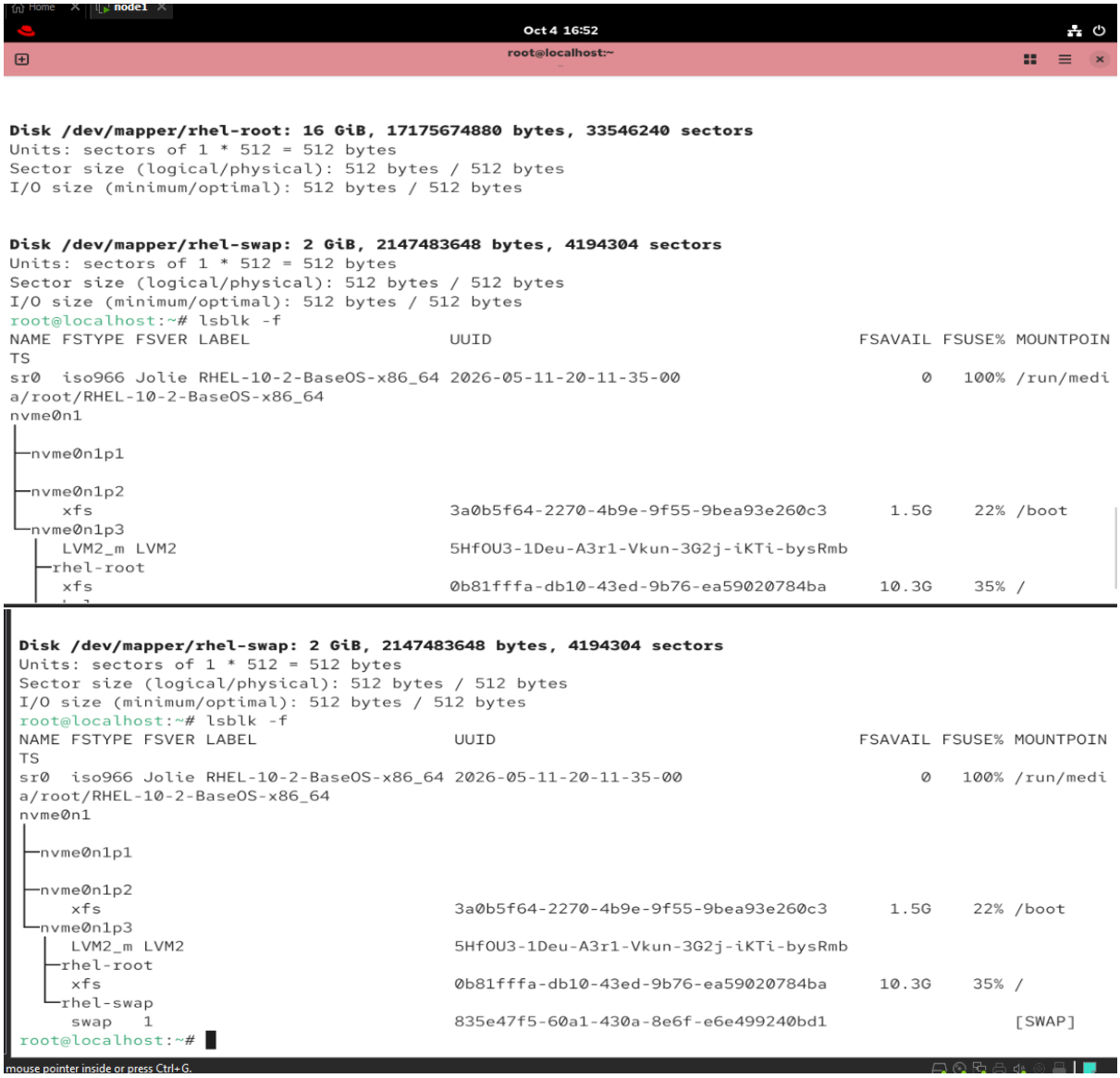
A. Identify disks

lsblk

sudo fdisk -l

lsblk -f





B. MBR/GPT

Check partition table:

sudo fdisk -l /dev/sdb

For GPT information:

sudo parted /dev/sdb print

Do not use /dev/sda unless you have positively confirmed it is the disposable lab disk.

C. Create partition

Example:

sudo fdisk /dev/sdb

Inside fdisk, commands vary according to the desired partition table. Always verify the disk before writing changes.

D. Create XFS

After creating a disposable partition such as /dev/sdb1:

sudo mkfs.xfs /dev/sdb1

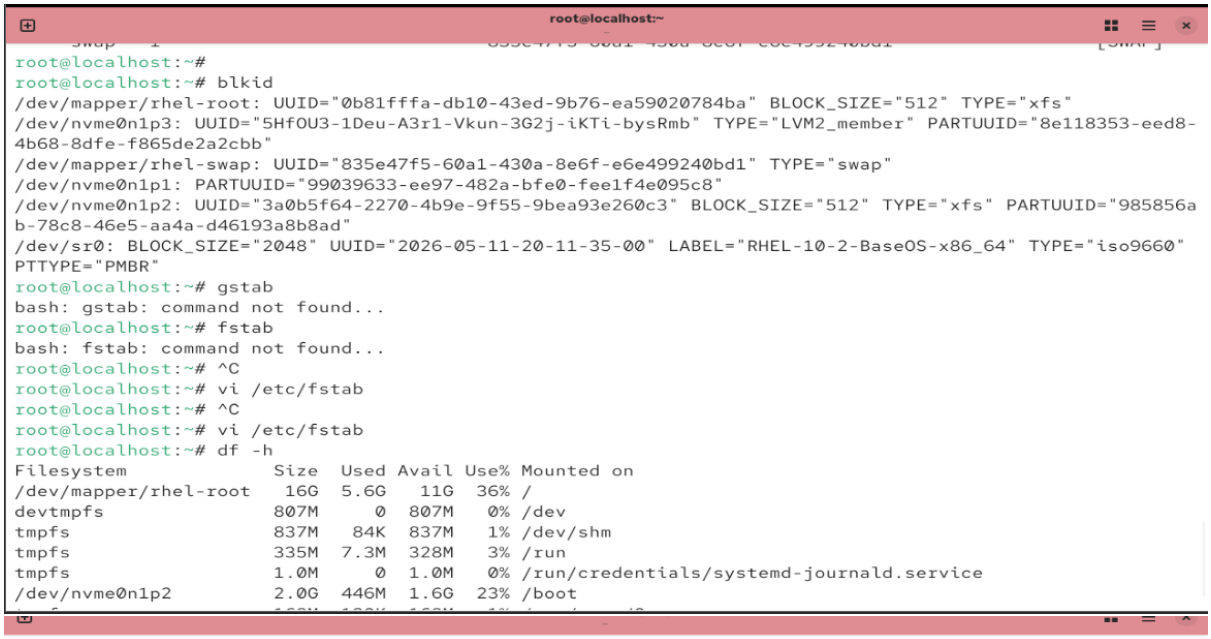
E. Create ext4
sudo mkfs.ext4 /dev/sdb1

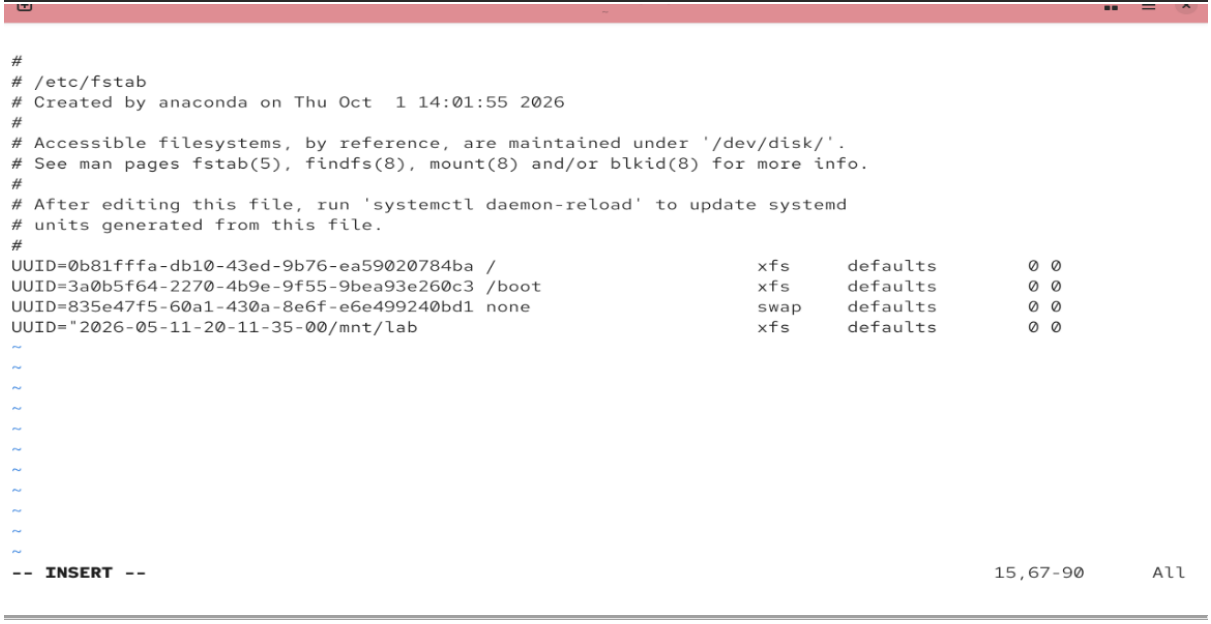
F. Create VFAT
sudo mkfs.vfat /dev/sdb1

G. Mount
Create mount point:
sudo mkdir /mnt/lab
Mount:
sudo mount /dev/sdb1 /mnt/lab
Verify:
df -h /mnt/lab
Unmount:
sudo umount /mnt/lab

H. UUID
lsblk -f
or:
sudo blkid /dev/sdb1
Example:
/dev/sdb1: UUID="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx" TYPE="xfs"

I. Mount using UUID
Edit:
sudo vi /etc/fstab
Example:
UUID=YOUR-UUID-HERE /mnt/lab xfs defaults 0 0
Test:
sudo mount -a
Then:
df -h
Students must use the **actual UUID produced by their own system**, not the example UUID.





J. LVM

Display:

sudo pvs

sudo vgs

sudo lvs

On a disposable partition:

sudo pvcreate /dev/sdb1

Create volume group:

sudo vgcreate tyitvg /dev/sdb1

Create logical volume:

sudo lvcreate -L 1G -n tyitlv tyitvg

Create file system:

sudo mkfs.xfs /dev/tyitvg/tyitlv

Mount:

sudo mkdir /mnt/tyit

sudo mount /dev/tyitvg/tyitlv /mnt/tyit
Verify:
df -h /mnt/tyit

K. Swap
Check:
swapon --show
For a **dedicated disposable swap partition**, use the appropriate partition device:
sudo mkswap /dev/sdb2
sudo swapon /dev/sdb2
Verify:
swapon --show
Do not run mkswap on an existing system partition.

PRACTICAL 12
Network Configuration

Aim
To configure and verify IPv4/IPv6 addresses, interfaces, DNS and hostname using nmcli and nmtui.

A. Show interfaces
ip addr
ip link

B. NetworkManager
systemctl status NetworkManager

C. nmcli
Show connections:
nmcli connection show
Show devices:
nmcli device status
Show IP information:
nmcli device show

D. Configure IPv4
First identify the connection:
nmcli connection show
Example:
sudo nmcli connection modify "Wired connection 1" ipv4.addresses 192.168.56.20/24
Gateway:
sudo nmcli connection modify "Wired connection 1" ipv4.gateway 192.168.56.1
DNS:
sudo nmcli connection modify "Wired connection 1" ipv4.dns "8.8.8.8"
Set manual addressing:
sudo nmcli connection modify "Wired connection 1" ipv4.method manual
Activate:
sudo nmcli connection up "Wired connection 1"
The exact connection name and address range depend on the student's VirtualBox/VMware network configuration.

E. nmtui
sudo nmtui
Use:
Edit a connection
Configure IPv4/IPv6.

F. Hostname
hostnamectl
Set:

sudo hostnamectl set-hostname rhel-lab
Verify:
hostnamectl

G. Test network
ping -c 4 8.8.8.8
DNS test:
ping -c 4 redhat.com

PRACTICAL 13
SSH Configuration

Aim
To configure an SSH server and access a remote RHEL system using password and key-based authentication.

A. Install SSH server
sudo dnf install openssh-server
Start:
sudo systemctl enable --now sshd
Check:
systemctl status sshd

B. Find IP
ip addr

C. SSH Login
From another Linux machine:
ssh username@IP_ADDRESS
Example:
ssh labuser@192.168.56.20

D. Generate SSH Key
On the client:
ssh-keygen -t ed25519
When prompted, enter a **passphrase**.

E. Copy Public Key
ssh-copy-id labuser@192.168.56.20
Login:
ssh labuser@192.168.56.20
The private key should remain protected and should never be copied to the server.

PRACTICAL 14
Apache Web Server

Aim
To configure a basic Apache web server and create a virtual host.

Install Apache
sudo dnf install httpd
Start:
sudo systemctl enable --now httpd
Check:
systemctl status httpd

Create Web Page
sudo vi /var/www/html/index.html
Enter:

```
<!DOCTYPE html>
<html>
<head>
   <title>TYIT Linux Practical</title>
</head>
<body>
   <h1>Welcome to TYIT RHEL Web Server</h1>
   <p>Apache Web Server is working successfully.</p>
</body>
</html>
```

Test locally:
curl http://localhost

Firewall
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
Check:
sudo firewall-cmd --list-services

Apache Configuration Files
Important locations:
/etc/httpd/
/etc/httpd/conf/httpd.conf
/etc/httpd/conf.d/
/var/www/html/
/var/log/httpd/
Check:
httpd -t
Expected:
Syntax OK

Virtual Host

Create:

sudo vi /etc/httpd/conf.d/tyit.conf

Example:

```
<VirtualHost *:80>
    ServerName tyit.local
    DocumentRoot /var/www/tyit

    <Directory /var/www/tyit>
        AllowOverride None
        Require all granted
    </Directory>
</VirtualHost>
```

Create directory:

sudo mkdir -p /var/www/tyit

Create page:

echo "TYIT Virtual Host" | sudo tee /var/www/tyit/index.html

Check configuration:

sudo httpd -t

Restart:

sudo systemctl restart httpd

PRACTICAL 15
Firewall Configuration

Aim
To configure Linux firewall using firewalld and study the relationship with iptables/nftables.

A. Check firewalld
sudo systemctl status firewalld

B. Start firewall
sudo systemctl enable --now firewalld

C. Check active zone
sudo firewall-cmd --get-active-zones

D. List services
sudo firewall-cmd --list-services

E. Allow HTTP
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
Verify:
sudo firewall-cmd --list-services

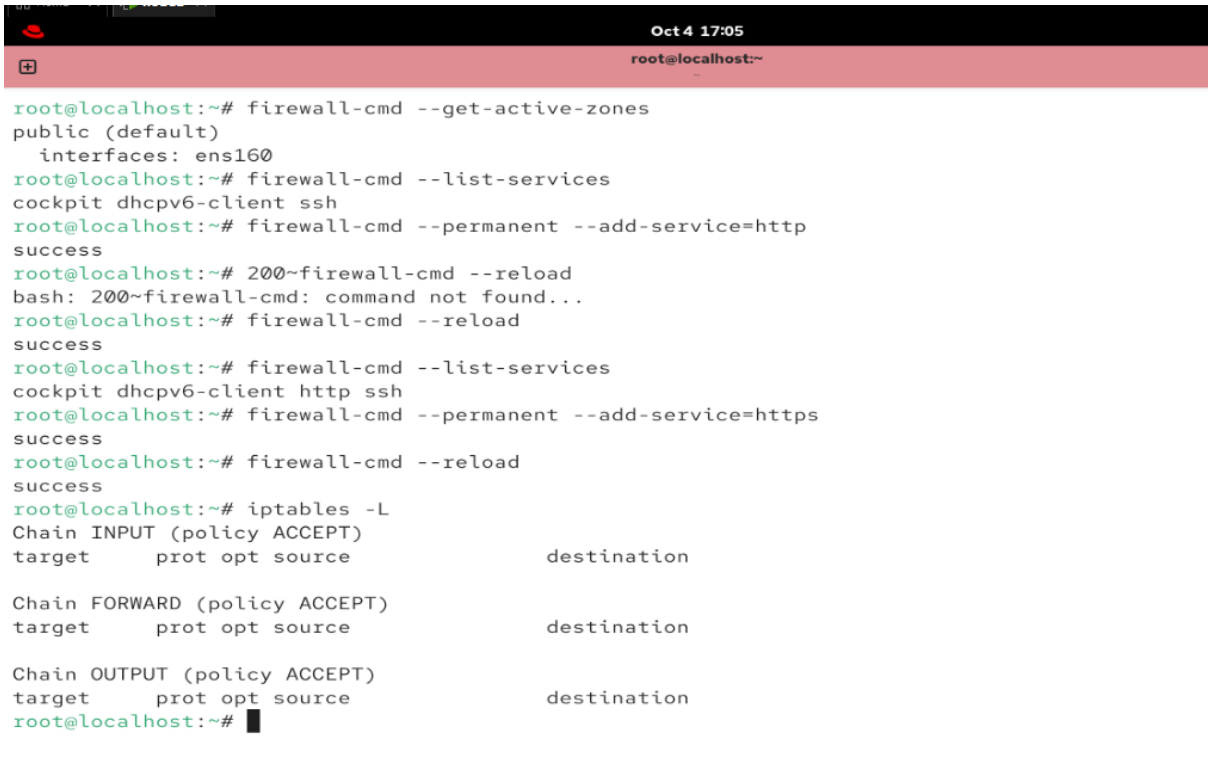
F. Allow HTTPS
sudo firewall-cmd --permanent --add-service=https
sudo firewall-cmd --reload

G. Remove service
sudo firewall-cmd --permanent --remove-service=http
sudo firewall-cmd --reload
RHEL documentation recommends predefined firewalld services for allowing required service traffic.

---

iptables
Check availability:
sudo iptables -L
On modern RHEL, students should understand that the underlying packet-filtering architecture is based around nftables, while firewalld provides the normal high-level management interface.


root@localhost:~
tmpfs 168M 120K 168M 1% /run/user/0
/dev/sr0 11G 11G 0 100% /run/media/root/RHEL-10-2-BaseOS-x86_64
root@localhost:~#
root@localhost:~# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
Active: active (running) since Sun 2026-10-04 16:33:45 IST; 27min ago
Invocation: ec2a832071674891a7204d9812e42f17
Docs: man:firewalld(1)
Main PID: 1212 (firewalld)
Tasks: 2 (limit: 10316)
Memory: 43.6M (peak: 48.5M)
CPU: 841ms
CGroup: /system.slice/firewalld.service
└─1212 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid
Oct 04 16:33:43 localhost systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon
Oct 04 16:33:45 localhost systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
root@localhost:~# systemctl enable --now firewalld
root@localhost:~# firewall-cmd --get-active-zones
public (default)
interfaces: ens160
root@localhost:~# firewall-cmd --list-services
cockpit dhcpv6-client ssh
root@localhost:~# firewall-cmd --permanent --add-service=http
success


Oct 4 17:05
root@localhost:~
root@localhost:~# firewall-cmd --get-active-zones
public (default)
interfaces: ens160
root@localhost:~# firewall-cmd --list-services
cockpit dhcpv6-client ssh
root@localhost:~# firewall-cmd --permanent --add-service=http
success
root@localhost:~# 200~firewall-cmd --reload
bash: 200~firewall-cmd: command not found...
root@localhost:~# firewall-cmd --reload
success
root@localhost:~# firewall-cmd --list-services
cockpit dhcpv6-client http ssh
root@localhost:~# firewall-cmd --permanent --add-service=https
success
root@localhost:~# firewall-cmd --reload
success
root@localhost:~# iptables -L
Chain INPUT (policy ACCEPT)
target prot opt source destination
Chain FORWARD (policy ACCEPT)
target prot opt source destination
Chain OUTPUT (policy ACCEPT)
target prot opt source destination
root@localhost:~#

PRACTICAL 16
NFS and CIFS Network Storage

Aim
To access remote NFS/CIFS network storage and configure persistent mounts.

A. NFS Client
Install:
sudo dnf install nfs-utils
List exports from server:
showmount -e SERVER_IP
Mount:
sudo mkdir /mnt/nfs
sudo mount SERVER_IP:/shared /mnt/nfs
Check:
df -h /mnt/nfs

B. NFS in /etc/fstab
Example:
SERVER_IP:/shared /mnt/nfs nfs defaults,_netdev 0 0
Test:
sudo mount -a

C. CIFS
Install:
sudo dnf install cifs-utils
Create mount point:
sudo mkdir /mnt/cifs
Mount:
sudo mount -t cifs //SERVER_IP/share /mnt/cifs -o username=USER
The command will prompt for a password where appropriate.

D. Automount

sudo dnf install autofs
Enable:
sudo systemctl enable --now autofs
Configuration depends on the NFS/CIFS server and mount map being used.

PRACTICAL 17
Shell Scripting

Aim
To create shell scripts for system information, automation and process monitoring.

A. System Information Script
Create:
vi systeminfo.sh
Enter:

```
#!/bin/bash

echo "===== SYSTEM INFORMATION ====="

echo "Hostname:"
hostname

echo "Kernel:"
uname -r

echo "Operating System:"
cat /etc/os-release | grep '^PRETTY_NAME'

echo "Current User:"
whoami

echo "Date:"
date

echo "Uptime:"
uptime

echo "Memory:"
free -h

echo "Disk:"
df -h /
```

Make executable:
chmod +x systeminfo.sh
Run:
./systeminfo.sh

B. Process Monitoring Script
Create:
vi process_monitor.sh
Enter:

```
#!/bin/bash
```

```
echo "===== TOP CPU PROCESSES ====="

ps -eo pid,user,%cpu,%mem,comm --sort=-%cpu | head -n 6

echo
echo "===== TOP MEMORY PROCESSES ====="

ps -eo pid,user,%cpu,%mem,comm --sort=-%mem | head -n 6
```

Run:

```
chmod +x process_monitor.sh
./process_monitor.sh
```

PRACTICAL 18
Scheduling Tasks Using Cron and At

Aim
To schedule recurring and one-time tasks using cron and at.

A. Cron
Check service:
systemctl status crond

Cron format
Minute Hour Day Month Weekday Command
Example:
* * * * * command
means every minute.

Create Cron Job
Edit:
crontab -e
Example:
*/5 * * * * date >> /home/student/cron-output.txt
This writes the date every five minutes.
List cron jobs:
crontab -l

B. at
Install if required:
sudo dnf install at
Enable service:
sudo systemctl enable --now atd
Schedule command:
echo "date >> /home/student/at-output.txt" | at 23:00
List jobs:
atq
Remove a job:
atrm JOB_NUMBER
